

RareAgentBench: A Multi-Pillar, Contamination-Audited Benchmark of LLM Agents for Rare-Disease Diagnosis

Anonymous ACL submission

Abstract

Rare disease diagnostic AI agents have proliferated (8+ systems in 2024-2026), yet no shared benchmark exists; each agent paper evaluates on an ad-hoc subset, making cross-system claims unverifiable. We introduce RareAgentBench, an agent-native benchmark spanning five capability pillars (phenotype extraction, phenotype-only DDx, genotype-aware DDx, family-aware DDx, clinical-communication faithfulness) on a layered dataset (Phenopacket-Store $n=10,051$; RareBench HF 1,122; RareArena RDS 72,661; post-cutoff PMC OA holdout $n=200$), plus a separately specified, exploratory MIMIC-IV structured-EHR protocol (956 code-supervised admissions; no clinical notes; replacement scoring pending). We implement eight published systems (five scaffolded diagnostic agents, RDMA extraction, and two classical/offline baselines) plus a no-scaffold LLM control, using four hosted backbones where compatible. H1-H11 and A1-A12 form a repository-defined analysis family; the OSF document remains an unregistered draft, so we do not claim formal pre-registration.

Key findings: (1) classical/offline baselines (LIRICAL 0.47, VC-RDAgent 0.44 R@1) decisively lead on Phenopacket-Store (best LLM cell 0.30, a 17 pp gap), while RareBench is near parity (best LLM 0.30 vs classical/offline 0.28); (2) multi-agent scaffolding gives only a small, dataset-dependent gain ($\approx 0-2$ pp R@1 over single-LLM controls), not the uniform boost prior work implies; (3) DeepSeek V4-Flash is $\sim 10\times$ cheaper than Gemini Flash but usually trades off accuracy (3-8 pp on Phenopacket-Store and 4-14 pp on RareArena, with mixed RareBench results) — cost-efficient, not quality-equivalent; (4) GPT-5 with `reasoning_effort=minimal` carries the highest receipt-weighted cost ($\sim 24\times$ V4-Flash) without a consistent accuracy edge — near-competitive on some scaffolds yet falling by ~ 9 pp on AgentClinic,

illustrating frontier-reasoning models’ brittleness under reasoning-disabled regimes; (5) a structured genotype channel adds about 20 pp R@1 in the paired HPO-plus-variant experiment, while the faithfulness-accuracy threshold remains judge-sensitive and exploratory. We release the harness, canonical case schema, per-agent adapter shims, full per-cell receipts, and a static-site leaderboard.

2 Introduction

2.1 The phenomenon — agent proliferation in rare disease

Rare disease diagnosis is undergoing a diagnostic-agent renaissance. In 2024–2026 alone, eight distinct agent systems were proposed that attack the problem with markedly different architectures: medical multi-agent debate (MDAgents, MedAgents, MAI-DxO) (Kim et al., 2024; Tang et al., 2024; Nori et al., 2025), OSCE dialogue simulation (AgentClinic) (Schmidgall et al., 2024), domain-specialised retrieval and reasoning (DeepRare) (Zhao et al., 2026), HPO-conditional fusion (VC-RDAgent) (cloudna-AI4LS, 2025), phrase-mining (RDMA) (Wu et al., 2025), and classical Bayesian likelihood (LIRICAL) (Robinson et al., 2020). Each system reports R@1 in the 0.3–0.7 range on its own evaluation set.

2.2 The gap — no shared benchmark

But no shared benchmark exists. Each paper evaluates on an ad-hoc subset: MDAgents on MedQA-Rare, MAI-DxO on NEJM clinicopathologic cases, DeepRare on a 9-source proprietary mix, RareBench on a 1.1k HPO subset, etc. Cross-system comparison is impossible. A recent systematic review and meta-analysis identified 15 studies contributing 19 system–dataset entries; every entry was assessed as high risk of bias, most often because of potential leakage and limited reproducibility, and benchmark disease composition

was associated with performance (Nguyen et al., 2026). Without a shared evaluation infrastructure, “agent X beats agent Y” is unverifiable.

2.3 Why agent-native, not LLM++

One could argue an LLM-only benchmark suffices. We disagree on three grounds. First, the diagnostic agent designs we audit differ on *non-LLM* axes — RAG pipelines (DeepRare), panel orchestration (MAI-DxO), OSCE dialogue (Agent-Clinic), classical fusion (VC-RDAgent) — that LLM-only benchmarks cannot disentangle from backbone choice. Second, the *evaluation surface* matters: pure accuracy ignores faithfulness, cost, and pass-k reliability, all of which matter in clinical deployment. Third, classical / offline baselines (LIRICAL Bayesian, VC-RDAgent Stage 1) are competitive with LLM agents on HPO-input datasets — a result that *no* prior benchmark would have surfaced because none of them run classical baselines side-by-side with the LLM lineup.

2.4 Our approach

We introduce RareAgentBench, a benchmark with three structural commitments:

Five capability pillars: phenotype extraction (P1), phenotype-only DDx (P2), genotype-aware DDx (P3), family-aware DDx (P4, deferred to v2), and clinical-communication faithfulness (P5). Each pillar surfaces different agent capabilities; collapsing all into a single accuracy number can hide faithfulness failures — a correct diagnosis can rest on an unfaithful reasoning trace — though we report the strength of that faithfulness/accuracy decoupling (Section B.3) as judge-dependent and exploratory rather than a firm quantitative claim.

Layered dataset: Phenopacket-Store (10,051 cases) → RareBench HF (1,122) → RareArena RDS (72,661) → PMC OA post-cutoff holdout (200). A separate, exploratory MIMIC-IV protocol defines early structured-event prediction plus ICD leakage auditing on 956 code-supervised admissions; replacement scoring is pending and is not pooled with the diagnostic layers. The layering spans difficulty, contamination risk, input modality (HPO vs free-text), and prevalence distribution. The PMC layer is publication-date held out (2024+), but an exact PMCID overlap audit prevents us from claiming that it eliminates contamination; we use it as a bounded temporal sensitivity analysis.

Repository-defined hypotheses + ablations:

H1–H11 + A1–A12 and the Holm correction are documented in the released analysis plan. The OSF file is still an unregistered draft with placeholder metadata, so the current study does not claim formal pre-registration.

2.5 Key findings preview

We implement eight published systems plus one no-scaffold control. Compatible LLM agents are evaluated across Gemini 3 Flash, DeepSeek V4-Flash, DeepSeek V4-Pro reasoning-off, and GPT-5 minimal; Table 1 reports the actual per-cell attempted N and Appendix J reports \$270.74 aggregate diagnostic cost.

Five findings:

1. Classical/offline baselines decisively lead on Phenopacket-Store: LIRICAL (Bayesian) R@1 = 0.47 [0.45–0.49]; VC-RDAgent (offline IC+Poincaré) 0.44 [0.40–0.48]; best LLM cell 0.30 (MedAgents × Gemini, N=2000) — a 17 pp gap. RareBench is near parity, with best LLM 0.30 and best classical/offline 0.28.
2. Multi-agent scaffolding gives only a small, dataset-dependent gain (≈ 0 –2 pp R@1) over single-LLM controls, not the uniform boost prior work implies (medagents 0.30 vs llm_control 0.29, Phenopacket-Store / Gemini, N=2000; within overlapping CIs — and several scaffolds fall *below* the control, see Section B).
3. DeepSeek V4-Flash is $\sim 10\times$ cheaper than Gemini Flash but usually trades off accuracy (R@1 -3 to -8 pp on Phenopacket-Store and -4 to -14 pp on RareArena; RareBench is mixed) — cost-efficient, not quality-equivalent.
4. GPT-5 with reasoning_effort=minimal is the most expensive backbone ($\sim 24\times$ V4-Flash, receipt-weighted) with no consistent accuracy edge — near-competitive on MedAgents yet falling by ~ 9 pp on AgentClinic dialogue — exposing frontier reasoning models’ brittleness under reasoning-disabled regimes.
5. Variant channel adds ~ 20 pp R@1 to *any* agent that ingests structured variants (Phase 3.2 P3 pilot: llm_control 0.26 → 0.46, deep-rare 0.22 → 0.38) — *not* DeepRare-specific.

2.6 Contributions

- **RareAgentBench benchmark:** 5 pillars $\times$ 3 diagnostic layers plus one structured-EHR probe $\times$ 9 implementations $\times$ 4 hosted backbones where compatible, with code and receipts released.
- **Per-agent adapter shims** (3,485 LOC): unified CanonicalCase input contract + subprocess isolation, with per-baseline reproducibility documentation.
- **Reproducibility receipts:** per-cell run-id + OpenRouter request-id
 - dollar cost; 90,046 successful diagnostic attempts aggregated in the frozen cost receipt.
- **Repository-defined statistical protocol:** Holm–Bonferroni for H1–H11, bootstrap 95% CIs, and LLM-judge sensitivity analysis with matched-trace Gemini/Claude agreement.
- **Static-site leaderboard** for community ratchet.

3 Related Work

3.1 Rare Disease LLM Benchmarks

Existing benchmarks are LLM-centric, not agent-native. RareBench (Chen et al., 2024) establishes a multi-subset rare-disease diagnostic protocol with rank-based metrics. RareArena (Chen et al., 2026) scales to free-text case reports, while Phenopacket-Store and subsequent Phenopacket studies provide structured HPO-based evaluation (Danis et al., 2025; Rekerle et al., 2026; Chimirri et al., 2025). MIMIC-RD (Eiz AlDin et al., 2026) introduces a note-based MIMIC-IV rare-disease task; unlike that resource, our local MIMIC probe has no notes. These benchmarks primarily evaluate static input-to-output diagnosis and do not jointly expose interactive tool use, cost/latency, reasoning-trace evaluation, and pass- k reliability. A systematic review of 15 studies (19 system–dataset entries) found all 19 at high risk of bias and highlighted substantial benchmark-dependent heterogeneity (Nguyen et al., 2026).

3.2 Agent Benchmarks in Other Domains

Agent benchmarks elsewhere have matured beyond static evaluation; rare-disease benchmarks have not. τ -bench formalizes repeated-trial reliability for tool-using agents (Yao

et al., 2024); AgentBoard introduces partial-credit progress metrics (Ma et al., 2024); and SWE-bench makes end-to-end issue resolution the unit of evaluation (Jimenez et al., 2024). In medicine, MedAgentBench covers tool-augmented FHIR tasks (Jiang et al., 2025), while MedHELM extends scenario-by-metric evaluation (Patel et al., 2025). We adopt three corresponding patterns: process-aware evaluation, pass- k reliability, and explicit cost/latency accounting.

3.3 Rare-Disease and Medical Agent Systems

Eight systems are relevant to rare-disease diagnosis or transferable to it, but they do not share an evaluation matrix. DeepRare integrates more than 40 tools and evaluates heterogeneous text, HPO, and genetic inputs (Zhao et al., 2026). MAI-DxO coordinates a multi-role diagnostic panel (Nori et al., 2025); RDMA specializes in phenotype extraction (Wu et al., 2025); and VC-RDAgent provides an offline ontology-based path (cloudna-AI4LS, 2025). From general medicine, MDAgents adapts collaboration mode (Kim et al., 2024), MedAgents orchestrates role-playing experts (Tang et al., 2024), and AgentClinic evaluates simulated clinical dialogue (Schmidgall et al., 2024). LIRICAL supplies a classical likelihood baseline (Robinson et al., 2020). Each was originally evaluated under a different input, dataset, or metric, leaving cross-system claims unverifiable on common ground.

We position this work as filling that exact gap.

4 Benchmark Design

4.1 Five Capability Pillars

We decompose rare-disease diagnosis into five distinct capability pillars, each individually testable, jointly forming the evaluation surface. The pillar choice is grounded in three considerations: (i) clinical workflow steps (phenotype recognition $\rightarrow$ DDx generation $\rightarrow$ genetic confirmation $\rightarrow$ family interpretation $\rightarrow$ communication), (ii) the dimensions on which existing rare-disease agents claim differentiation (DeepRare’s reference accuracy, MAI-DxO’s budgeted reasoning), and (iii) reviewer-anticipated criticism that accuracy alone is insufficient (Pfohl et al., 2024); Pillar 5 directly addresses this.

The five pillars, and which data layer carries each, are summarised in the benchmark-surface figure. Briefly: P1 Phenotype Extrac-

tion (free-text EHR $\rightarrow$ ranked HPO list; P/R/F1); P2 Phenotype-only DDx (HPO list $\rightarrow$ ranked diseases; Recall@1/3/5/10, median rank, MRR); P3 Genotype-aware DDx (HPO + structured variants $\rightarrow$ disease + causative gene; Gene Top-k, cross-mapped R@1); P4 Family-aware DDx (HPO + trio/pedigree $\rightarrow$ disease + mode of inheritance; deferred to v2 pending pedigree-bearing data); and P5 Reasoning Faithfulness (agent trace + prediction $\rightarrow$ a four-axis factual / relevance / depth / faithfulness score; the current 40-trace LLM-judge sensitivity study is exploratory, and physician validation remains future work).

Why five and not three. P1 and P5 in particular face the objection “isn’t this just preprocessing or just a stylistic axis?”. Both are load-bearing for our central claim. P1 is independent because Section E shows, in a same-case paired test, that downstream P2 R@1 is measurably sensitive to HPO-extraction quality — extraction is not a free preprocessing step (and the dual-pass design is what lets this be measured without confounding it with dataset difficulty). P5 is independent because Section B.3/Section 7 examine whether faithfulness ranking decouples from accuracy ranking (repository-plan Hypothesis 10); we report H10 as exploratory (a modest same-trace judge difference remains, but the current Gemini-to-Claude comparison cannot identify a pure family effect), and the durable point is that accuracy-only evaluation can miss hallucinated citations and unfaithful reasoning chains — a Type-1 error category discussed by DeepRare (Zhao et al., 2026).

Bias as cross-cutting lens, not pillar. Following HELM and MedHELM precedent (Liang et al., 2022; Patel et al., 2025), we apply available stratifications (prevalence tier, specialty, age group, sex, and HPO density) across relevant pillar metrics rather than treating bias as a separate axis. Dedicated fairness evaluations (Pfohl et al., 2024; Schaekermann et al., 2024; Omiye et al., 2023; Zack et al., 2024) remain complementary: ancestry and language analyses cannot be claimed in v1 because the current frozen resources do not support them.

4.2 Datasets — Three Diagnostic Layers + One Structured-EHR Probe

We assemble three diagnostic layers and one secondary structured-EHR probe; their sources and sizes appear as the columns of the benchmark-surface figure, and the full per-layer breakdown

(disease counts, ID anchors, free-text / gold-HPO / variant availability) is detailed in Appendix A. The diagnostic layers are L1 Phenotype Backbone (Phenopacket-Store + RareBench HF; 11,173 cases; HPO-only, gold HPO, variants on PP-Store), L2 Scale + Free Text (RareArena RDS/RDC; 72,661 verbatim case reports), and L4 Cutoff-After Holdout (self-built PMC OA, publication date $\geq$ 2024-01-01; 198 model-verified cases with physician annotation in progress). Separately, **S-EHR** is a credentialed MIMIC-IV-3.1 probe (Johnson et al., 2024) (956 admissions, 239 exact-mapped ORPHA labels) for early structured-event prediction and ICD leakage auditing. The replacement protocol is specified but not yet scored; it is not pooled with differential-diagnosis results.

Rationale per layer.

- **L1** establishes apples-to-apples comparison with RareBench’s KDD’24 numbers — required by reviewers (“why not just compare to RareBench”).
- **S-EHR** is designed to test whether agents can use real structured hospital events rather than curated case reports. The 24-hour primary snapshot uses timestamped labs, medications, procedures and services; it excludes target-bearing diagnosis codes/titles, post-window events and free text. Gold is derived from exact ICD-10 $\rightarrow$ Orphanet mapping, so this is code-supervised retrospective prediction, not independently adjudicated diagnosis. A paired audit will compare ICD titles, ICD codes only, and context after removing target-bearing entries.
- **L2** addresses scale; RareArena’s 72,661 cases span 45.6% of Orphanet. Released CC-BY-NC-SA, so we use it for academic evaluation only and acknowledge license bounds.
- **L4** responds to the leakage and benchmark-composition concerns in the 2026 systematic review (Nguyen et al., 2026). We extract PMC OA case reports published after 2024-01-01 via E-utilities and map diagnoses to Orphanet. The 198-case post-cutoff set has model-based verification but physician annotation is still in progress. Exact PMCID overlap RareArena, so L4 is a temporal sensitivity analysis, not an independently clean holdout.

Total resource pool: ~85,000 cases, with v1

analyses stratified where supported by prevalence tier, specialty, age group, sex, and HPO density. The current scored diagnostic study is English-language; a Chinese diagnostic layer is deferred (Section 8).

Evaluation N per dataset — honest disclosure. We deliberately separate “pool size” (the released benchmark) from “evaluation N” (what we run for the v1 paper). - **RareBench-HF** has 1,122 cases. Primary general-agent cells use the full attempted N; DeepRare and MAI-DxO have explicitly smaller adapter-specific N. - **Large layers** (Phenopacket-Store 10,051; RareArena RDS 72,661): the primary `llm_control/MDAgents/MedAgents` matrix uses a shared, prevalence-stratified cap of N=2,000 case IDs (seed=42). DeepRare, MAI-DxO, and a small number of failed-to-launch adapter cells have smaller attempted N shown in Table 1. - **MIMIC** will use its own attempted-admission denominator and separate receipts after the replacement experiment is run.

This is a prevalence-stratified evaluation, not a power-stratified extrapolation to full pool; bootstrap CIs in Section 6 quantify the resulting uncertainty per cell. We confirmed via our sampling-validation check that the N=2,000 stratified sample reproduces the full-N prevalence band and HPO-organ-system distribution within ± 2 pp.

4.3 Canonical Case Representation

The three diagnostic layers ingest into a single Pydantic v2 schema, `CanonicalCase` (illustrated in the canonical-case figure), and every agent adapter projects from this representation to the agent’s native input. The pending MIMIC protocol uses an analogous model-input/evaluation-only record and maps the structured snapshot at the adapter boundary. The record groups an identity block (`case_id`, `source_dataset`, `source_split`, `language`), an input block (demographics; `free_text_vignette` / `synthetic_vignette`; `gold_hpo_terms`; `variants`; local `vcf_path`; `family`), a parallel-ID gold label, and free-form metadata.

Three design decisions deserve note: (a) gold labels are parallel IDs (OMIM/ORPHA/CCRD), reflecting genuine ontology disagreement across datasets — Phenopacket-Store uses OMIM, RareArena uses ORPHA, CCRD anchors the Chinese-listed 207 diseases. Evaluator must ac-

cept cross-mapped matches via Orphadata (Section 4.5). (b) `synthetic_vignette` is distinguished from `free_text_vignette` so we can audit any evaluation that relies on LLM-synthesized prose (Section B disclosure). (c) `vcf_path` carries a local-only file pointer; PhysioNet DUA prohibits transmission of identifiable EHR data to external LLM APIs, so adapter shims projecting Pillar 3 inputs convert structured variant info to abstracted strings before any cloud-LLM call.

4.4 Dual-pass evaluation

Every pillar is evaluated in two modes. In Pass A (gold-HPO, primary) the case’s curated HPO terms are fed directly to the agent’s downstream pillar, bypassing its own extractor — isolating downstream capability and enabling apples-to-apples comparison on identical inputs. In Pass B (end-to-end) the raw free-text vignette is fed in and the agent extracts HPO itself before reasoning, measuring real deployment performance. The Pass A — Pass B delta is itself a reported metric (following RareBench’s phenotype-vs-EHR-text comparison (Chen et al., 2024)); we show it is non-uniform across agents (Section E), which turns the input heterogeneity of a mixed-agent lineup into a measured axis rather than a confound (Section D).

4.5 Metrics and matching

We report Recall@1/3/5/10, median rank and MRR for the DDx pillars, P/R/F1 for phenotype extraction, and task-success rate; a recommended tier (pass[^]k, cost-normalised accuracy, calibration, reference accuracy) and an exploratory tier (step-level and reasoning-faithfulness scores) are defined in Appendix C. Every accuracy metric is stratified where metadata and sample size permit; v1 reports prevalence tier, specialty, age group, sex, and HPO-density analyses, while ancestry and language remain deferred. A predicted disease ID counts as a hit if it is prefix-equal on the same ontology, cross-mapped via Orphadata (OMIM ↔ ORPHA), or fuzzy-matches an Orphanet name or synonym at rapidfuzz score ≥ 90 — a threshold calibrated against 217 audited borderline cases (>85% of the 70–89 band were false positives); the audit tape is released for reproducibility (Appendix N).

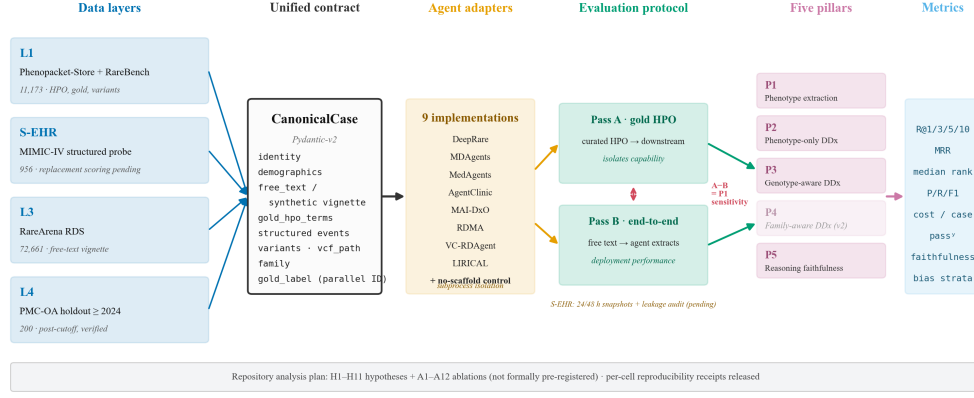


Figure 1: **RareAgentBench overview.** The three diagnostic layers use a shared `CanonicalCase` contract; the separately reported MIMIC replacement probe preserves the same model-input/gold separation and is projected at the adapter boundary. Eight published systems plus one no-scaffold control run through subprocess-isolated shims. Diagnostic pillars use the two-pass protocol; the pending MIMIC experiment is specified to use timestamped structured events and an ICD leakage audit.

5 5.2-5.4 Experimental Setup — Backbones / Adapter Methodology / Analysis Plan

5.1 Backbones

We evaluate every LLM-driven agent against four backbones spanning the cost–capability frontier: DeepSeek V4-Flash (open-weight, low-cost), DeepSeek V4-Pro (open-weight frontier, reasoning disabled), Gemini 3 Flash (our primary baseline), and GPT-5 (frontier, minimal reasoning). All are accessed through OpenRouter for version-pinned endpoints and a single billing and logging surface. Their dated aliases, pricing, context windows and the generation settings held constant across every cell are listed in Appendix Table 5.

All four backbones are evaluated in their minimal/off reasoning configuration (Gemini thinking-off, GPT-5 `reasoning_effort=minimal`, DeepSeek-V4-Pro reasoning disabled, V4-Flash light-reasoning within budget). This is a deliberate design choice, not an artifact: (i) it isolates the scaffolding contribution — the benchmark’s object of study is whether agent scaffolds help, so the backbone’s internal chain-of-thought is held to a constant floor rather than left to confound scaffold-vs-backbone-CoT gains; (ii) it matches the original operating point of the reproduced agents (MDAgents/MedAgents/AgentClinic were published on non-reasoning GPT-3.5/GPT-4); and (iii) it is required for tractability at $N=10^3$ cases $\times$ multi-call scaffolds. We report reasoning-on separately as a thinking-mode ablation (H6 / Ablation A8,

Section C) on the single-call LLM control.

Methods note 1 (GPT-5, reviewer-defensive). GPT-5’s default `reasoning_effort=high` silently routes the full `max_tokens` budget into hidden reasoning tokens, returning empty content. In our first Phase 2 GPT-5 run this manifested catastrophically: 50/50 MedAgents `parser_error` (raw response empty), 50/50 AgentClinic timeout, 46/50 MAI-DxO timeout. We force `reasoning_effort=minimal` for all primary results, propagating the flag through subprocess-isolated adapters via the `OPENROUTER_REASONING_EFFORT` environment variable. MAI-DxO + GPT-5 remained incompatible even at `minimal` (panel orchestration $\times$ `max_iter=3` exceeds the subprocess cap); we document the incompat in Section 8 Limitations and exclude that single cell, not the GPT-5 row.

Methods note 2 (DeepSeek-V4-Pro, reviewer-defensive). V4-Pro is a heavy reasoning model whose reasoning is *unbounded* and, unlike GPT-5, ignores every throttle knob — verified on a hard prompt ($N \geq 3$ each): `reasoning_effort=minimal` and `reasoning_effort=low` are both ignored (reasoning still fills the budget); capping via `reasoning={max_tokens:200}` is ignored; and simply enlarging `max_tokens` is self-defeating because reasoning scales to consume whatever it is given (at `max_tokens=2500` the synthesiser emitted content in 0/4 trials; at 4000, 3/4 but one trial still consumed all 4000). The subprocess baselines size `max_tokens` for non-

reasoning models (AgentClinic doctor turn 200; MedAgents synthesiser 600), so V4-Pro’s reasoning consumed the entire budget and returned empty content, surfacing as AgentClinic retry-loop timeouts (45–51% of cells) and MedAgents `parser_error` (34–45%). The only lever that actually disables V4-Pro reasoning is `reasoning={"enabled": false}` (verified: 0 reasoning tokens, content in 3/3 trials, 1.9 s vs 32 s). We propagate this via `OPENROUTER_REASONING_DISABLE` through the same subprocess-env mechanism. This puts V4-Pro in the same reasoning-off configuration as GPT-5-minimal, restoring cross-backbone consistency; it also cut AgentClinic wall-clock from >900 s to ~27 s/case (33×). Full root-cause receipts are recorded in our run worklog and the per-baseline reproduction notes (Appendix B).

5.2 Per-Agent Adapter Shim Methodology

Each of the eight agents ships its own Python environment, often pinned to an `openai < 1.0` SDK and incompatible with one another in the same process. We isolate each adapter behind a `subprocess.run` boundary into its own `virtualenv`, with a uniform Python-side `AgentAdapter` abstract base class that takes a `CanonicalCase` (Section 4.3), projects it into the agent’s native input format (MCQA prompt, OSCE scenario, phenopacket JSON, ...), invokes the subprocess, and parses `stdout` into a uniform `PredictionLog` (ranked candidates, extracted HPO, latency, cost, reasoning trace, status).

Backbone wiring is never modified in the agent’s core logic; we only patch the `openai` client construction sites to accept an `OpenRouter` `base_url` and propagate `OPENROUTER_API_KEY` / `OPENROUTER_REASONING_EFFORT` / `OPENROUTER_REASONING_DISABLE` through the subprocess env (the latter two forward `reasoning={effort}` and `reasoning={enabled:false}` respectively; see Methods notes 1–2). Patches range from 3 LOC in DeepRare’s API interface to ~30 LOC for AgentClinic’s `--openrouter` CLI flag, and are enumerated in the Agent Fairness Matrix (Section D).

What this gives us. (i) Per-agent reproducibility — every run report contains the verbatim subprocess invocation. (ii) Compositional cost accounting — our `OpenRouter` logging wrapper captures token usage and dollar cost per call, propagated up through the subprocess via a JSONL log

file. (iii) Failure isolation — one agent’s RAG / panel / parser bug cannot poison another agent’s evaluation.

Caveats we surface. (a) Subprocess isolation costs ~0.5–2 s wall-clock overhead per case, dominated by interpreter startup; we report adapter overhead alongside agent latency in the experimental-settings table. (b) Cost reporting is exact for adapters using our wrapper (`mdagents`, `medagents`, `agentclinic`, `deeparare`, `maidxo`, `llm_control`) and estimated from tokens for adapters whose upstream code bypasses the wrapper (`rdma`, `lirical`, `vc_rdagent`). (c) Three adapters required defensive output-dir purging to prevent first-case state leak — DeepRare’s deterministic output filename was the most severe. All such patches are documented per-agent.

5.3 Repository Analysis Plan and Multiplicity Control

The repository enumerates hypotheses (H1–H11), ablations (A1–A12), the staged-sampling budget guard, Holm–Bonferroni multiple-testing correction, an LLM-judge sensitivity protocol, and a post-cutoff holdout procedure. These labels define the analysis family used throughout this paper and make additions and deviations auditable.

This is not a formal pre-registration claim. The bundled OSF document is an unregistered draft with a placeholder ID and date, and the public repository history does not independently establish that the plan preceded all reported analyses. We therefore describe H1–H11 and A1–A12 as repository-defined tests and ablations, not as prospectively registered confirmatory hypotheses. The Holm correction controls multiplicity within the six tests computable from the frozen receipts, but it does not convert a retrospective analysis into a pre-registered one.

Operational integrity. The PMC OA post-cutoff set is treated as a temporal sensitivity analysis, with exact source overlap disclosed in Section B.7 rather than described as an untouched or contamination-free holdout. Every diagnostic cell in Table 1 ships with a per-cell reproducibility receipt (run ID, `OpenRouter` request ID where available, and cost). Future confirmatory evaluation should register the protocol before collecting or unblinding a new, independently curated holdout.

6 Main Results

6.1 Headline R@1 Matrix(per-dataset, N in brackets)

Sorted by PP-Store R@1 (descending); classical/offline baselines listed first.

MIMIC-IV is not a column in this diagnostic matrix: its current input and code-derived outcome define a different structured-EHR task (Section 4.2), reported separately after the replacement protocol is scored and audited.

Notes: Bracketed N is attempted N, matching the R@1 denominator; completion counts and failure types remain in the released receipts. (1) *lirical*/*vc_rdagent* run only on HPO-input datasets (PP-Store / RareBench), so their Avg is not comparable to the 3-dataset LLM rows and is marked n/a (2-ds). (2) *maidxo* is systematically weak across all backbones (its panel degrades on HPO-list input, Section B) and *maidxo*×GPT-5 is incompatible (Section 8 L1); we list the Gemini row as representative. (3) *deeparare* DS V4-Pro on RareBench scores 0.22 [n=74], but n is too small (P3-only) to enter the main ranking (see Section B.2). (4) The DS V4-Pro column is reasoning-off (Section 5.1 Methods note 2); the thinking-mode contrast is in Section C H6.

Key cells (the primary *llm_control*/*mdagents*/*medagents* PP-Store and RareArena cells use the common N=2000 cap; smaller adapter-specific N are shown):

- Classical/offline baselines lead on PP-Store (*lirical* 0.47, *vc_rdagent* **0.44**), above any LLM row (best: *medagents* Gemini 0.30 / *llm_control* Gemini 0.29) by 17 pp (LIRICAL 0.47 – *medagents* 0.30; the gap is 18 pp against the single-LLM control at 0.29) — headline finding F1 is further strengthened on the unified large sample.
- On RareBench, *deeparare* (0.29-0.30) and the classical baselines (*lirical* 0.24 / *vc_rdagent* 0.28) lead, while every other LLM scores ≤ 0.10 — see the F5 ORPHA-sibling explanation.
- *deeparare* scores 0.00 on RareArena free text (a structural limitation; see the DeepRare reproduction note in Appendix B).
- **V4-Pro reasoning-off is not uniformly crippled**: it reaches 0.28 on *medagents* PP-Store, close to Gemini at 0.30, although other V4-Pro cells span 0.19–0.27 and Gem-

ini is the PP-Store winner for every listed agent. This is compatible with the separate Section C H6 result that enabling thinking mode on the single-call control was not cost-effective.

6.2 Backbone × scaffolding interaction

We hold the central backbone constant per agent and vary across {Gemini Flash, DS V4-Pro (reasoning-off), DS V4-Flash, GPT-5 minimal}. The four primary general-agent cells use the full attempted N; DeepRare’s per-backbone N is smaller and printed in the source receipts. Per-agent backbone winners (R@1 PP-Store):

Gemini Flash is the observed PP-Store winner for every listed agent in the frozen manifest, although its advantage over V4-Pro/GPT-5 is only 2–5 pp for the three main scaffolds and the single-call control. This is descriptive backbone sensitivity, not a clean backbone main effect: DeepRare cells have smaller and unequal N, and scaffold call patterns alter cost and failure rates. DS V4-Pro is evaluated reasoning-off (Section 5.1); its separate thinking-mode ablation adds ≈ 0 R@1 at 40% no-answer cost (Section C H6).

6.3 Cost-vs-Accuracy(per-prediction USD)

Cost per prediction spans more than an order of magnitude across backbones: DS V4-Flash is the cheapest hosted backbone (receipt-weighted mean \$0.00033/pred), DS V4-Pro reasoning-off averages \$0.00097/pred, and GPT-5 minimal is $\sim 24\times$ more expensive than V4-Flash with no consistent R@1 advantage (it is near V4-Pro on *medagents* and among the weakest on *agentclinic* — see F4). The cost-vs-accuracy trade-off across every agent × backbone cell is shown in the cost-vs-accuracy figure, and the full per-backbone and per-cell cost breakdown (diagnostic total \$270.74, 75% of the planned \$360 cap) is in Appendix J.

6.4 Headline Findings

V4-Pro re-run **reasoning-off** after root-causing an unbounded-reasoning starvation bug (Section 5.1 Methods note 2). (2) **PP-Store/RareArena harmonized to a > common N=2000 case-id cap** across all backbones for the primary *llm_control*/*mdagents*/*medagents* cells (earlier cells ranged 500–4589 due to historical over-runs, breaking cross-backbone comparability). Adapter-

Table 1: Headline Recall@1 (variant-aware) for every agent $\times$ backbone cell across the three diagnostic datasets; bracketed values are attempted N , which is also the R@1 denominator. The distinct MIMIC structured-EHR replacement is pending and will be reported separately. Cell shading encodes R@1: $<.10$.10–.20 .20–.30 .30–.40 $\geq .40$.

Agent	Backbone	PP-Store	RareArena	RareBench	Avg
lirical (classical)	—	0.47 [2000]	n/a HPO	0.24 [1122]	n/a (2-ds)
vc_rdagent (offline)	—	0.44 [663]	n/a HPO	0.28 [1122]	n/a (2-ds)
medagents	Gemini Flash	0.30 [2000]	0.30 [2000]	0.05 [1122]	0.22
llm_control	Gemini Flash	0.29 [2000]	0.28 [2000]	0.02 [1122]	0.20
deepar	Gemini Flash	0.28 [610]	0.00 [500]	0.30 [954]	0.19
medagents	DS V4-Pro	0.28 [2000]	0.23 [2000]	0.01 [1122]	0.17
mdagents	Gemini Flash	0.28 [2000]	0.28 [2000]	0.10 [1122]	0.22
medagents	GPT-5 min	0.28 [2000]	0.26 [2000]	0.01 [1122]	0.18
llm_control	DS V4-Pro	0.27 [2000]	0.19 [2000]	0.02 [1122]	0.16
mdagents	DS V4-Pro	0.27 [2000]	0.22 [2000]	0.04 [1122]	0.17
llm_control	DS V4-Flash	0.26 [2000]	0.21 [2000]	0.05 [1122]	0.17
llm_control	GPT-5 min	0.25 [2000]	0.22 [2000]	0.01 [1122]	0.16
medagents	DS V4-Flash	0.25 [2000]	0.16 [2000]	0.03 [1122]	0.15
mdagents	DS V4-Flash	0.25 [2000]	0.23 [2000]	0.05 [1122]	0.17
mdagents	GPT-5 min	0.24 [2000]	0.22 [2000]	0.01 [1122]	0.16
deepar	DS V4-Flash	0.22 [500]	0.00 [500]	0.29 [785]	0.17
agentclinic	Gemini Flash	0.21 [2000]	0.13 [2000]	0.01 [1122]	0.12
agentclinic	DS V4-Pro	0.19 [2000]	0.12 [2000]	0.01 [1122]	0.10
agentclinic	DS V4-Flash	0.13 [1955]	0.10 [1981]	0.02 [920]	0.08
agentclinic	GPT-5 min	0.13 [2000]	0.11 [2000]	0.00 [1122]	0.08
maidxo	Gemini Flash	0.02 [100]	0.06 [100]	0.00 [831]	0.03

Table 2: Per-agent Phenopacket-Store backbone sensitivity: best vs. worst R@1. Primary general-agent cells use attempted $N=2000$; DeepRare uses smaller unequal N , so its row is descriptive.

Agent	Best backbone	Worst backbone
llm_control	Gemini Flash (0.29)	GPT-5 min (0.25)
mdagents	Gemini Flash (0.28)	GPT-5 min (0.24)
medagents	Gemini Flash (0.30)	DS V4-Flash (0.25)
agentclinic	Gemini Flash (0.21)	GPT-5 min / DS V4-Flash (0.13)
deepar	Gemini Flash (0.28; n=610)	DS V4-Pro (0.20; n=111)

specific DeepRare/MAI-DxO cells remain smaller and are labelled. Net effect: R@1 estimates settled $\sim 2\text{--}4$ pp below the small-sample values (the 500-case samples were mildly optimistic), so F1 *widens* (17–18 pp) and F4’s “GPT-5 best for medagents” no longer holds (medagents Gemini 0.30 $\geq$ GPT-5 0.28). F2/F3 directions unchanged.

F1: Classical / offline leads decisively on Phenopacket-Store and at the super-rare prevalence tail. LIRICAL (Bayesian) R@1=0.47, VC-RDAgent Stage 1 (offline IC+Poincaré) R@1=0.44 on Phenopacket-Store (common $N=2000$ sample), against the best LLM cell (medagents $\times$ Gemini R@1=0.30; llm_control $\times$ Gemini 0.29) — a **17–18 pp** gap that *widened* under the larger harmonized sample (the earlier $N\approx 100\text{--}500$ optimistic estimates of 0.31–0.36 regressed to 0.29–0.30 at $N=2000$). RareBench does not reproduce a blanket classical advantage: DeepRare (0.29–0.30) slightly exceeds the best classical/offline result (VC-RDAgent 0.28), while all other LLM scaffolds

sit ≤ 0.10 . The robust cross-family result is therefore dataset- and prevalence-specific: the Phenopacket-Store gap above and H1’s +28 pp classical advantage on the super-rare tier, not a claim that classical wins every HPO-input dataset.

F2: Multi-agent scaffolding gives at most a small, dataset-dependent gain and often none, not a uniform boost. On Gemini Flash (common $N=2000$), medagents (PP-Store 0.30, RareArena 0.30) edges llm_control (0.29, 0.28) by only $\sim 1\text{--}2$ pp, within the control’s CI; mdagents (PP-Store 0.28) actually sits ~ 1 pp *below* the control, and agentclinic/maidxo regress sharply (Section B). The benefit does not consistently exceed the no-scaffold control’s CI and does not hold on every backbone. (Revised down from the v0 “+5–7 pp” claim, which rested on a stale small-sample medagents 0.40.)

F3: DeepSeek V4-Flash is $\sim 10\times$ cheaper than Gemini Flash but usually trades off accuracy. Receipt-weighted mean cost is \$0.00033 (V4-Flash) vs \$0.00344 (Gemini). On PP-Store, V4-Flash is 3–8 pp lower across the comparable agents; on RareArena the gap ranges 4–14

pp. RareBench is mixed, including cells where V4-Flash ties or exceeds Gemini, so the quality claim is not universal. V4-Flash also showed a higher transient empty-content rate on free-text/HPO-list inputs (mitigated by a wrapper-level retry; see Appendix B). Conclusion: V4-Flash is the cost-efficient choice when $\sim 10\times$ cost reduction outweighs the observed accuracy drop, but it does NOT match Gemini quality. (Reversed from v0.)

F4: GPT-5 minimal-reasoning is not worth its cost, and at full-N has no scaffold where it is the sole winner. GPT-5 (reasoning_effort=minimal, forced because default reasoning consumes all max_tokens) reaches 0.28 on medagents PP-Store (below Gemini at 0.30 and tied with V4-Pro-off) and falls on AgentClinic OSCE dialogue (0.13 versus Gemini 0.21, about -9 pp). As the most expensive backbone ($\sim 24\times$ V4-Flash per prediction) with no consistent accuracy edge, it is hard to justify. The reasoning-channel question is answered directly by our H6 ablation (Section C): on the single-call LLM control, turning reasoning on (V4-Pro) changes R@1 by $+0.008$ (noise) while producing no parseable answer in 40% of cases and running $10\text{--}40\times$ slower — thinking mode is not worth it on this task.

F5: RareBench HF is uniquely hard for general LLM scaffolds (≤ 0.10 R@1) but tractable for classical/HPO-pipeline agents (lirical 0.24, vc_rdagent 0.28, deeprare 0.29–0.30). Root causes: (a) RareBench gold labels use ORPHA codes with sibling-disambiguation challenges across Orphanet’s hierarchy (Methylmalonic acidemia with homocystinuria ORPHA:26 vs Vitamin B12- unresponsive methylmalonic acidemia ORPHA:27 share concept but not OMIM cross-ref); (b) classical/HPO agents use OMIM directly + Orphanet name fuzzy match, bypassing this mismatch. A real evaluator-vs-data interaction, not a pure model failure. Adapter-side fuzzy variants logging recovers $+1\text{--}8$ pp but doesn’t close the gap.

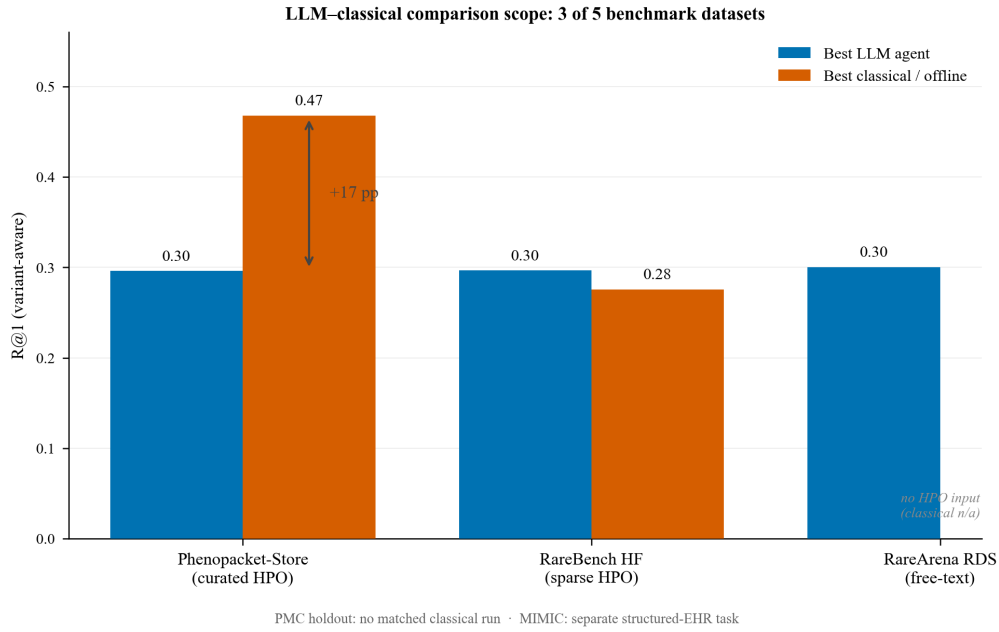


Figure 2: Scope of the LLM-classical comparison (F1). The benchmark has five dataset resources, but this panel covers the three datasets in the frozen headline diagnostic matrix. Phenopacket-Store and RareBench have paired LLM and classical/offline results; RareArena contributes only an LLM bar because the classical systems require HPO input. The PMC holdout has no matched classical run, and MIMIC is a distinct structured-EHR task, so neither belongs in this comparison. Values are variant-aware R@1 with the attempted denominator: Phenopacket-Store $N=2000$ per plotted bar; RareBench best LLM $N=954$ and classical $N=1122$; RareArena best LLM $N=2000$.

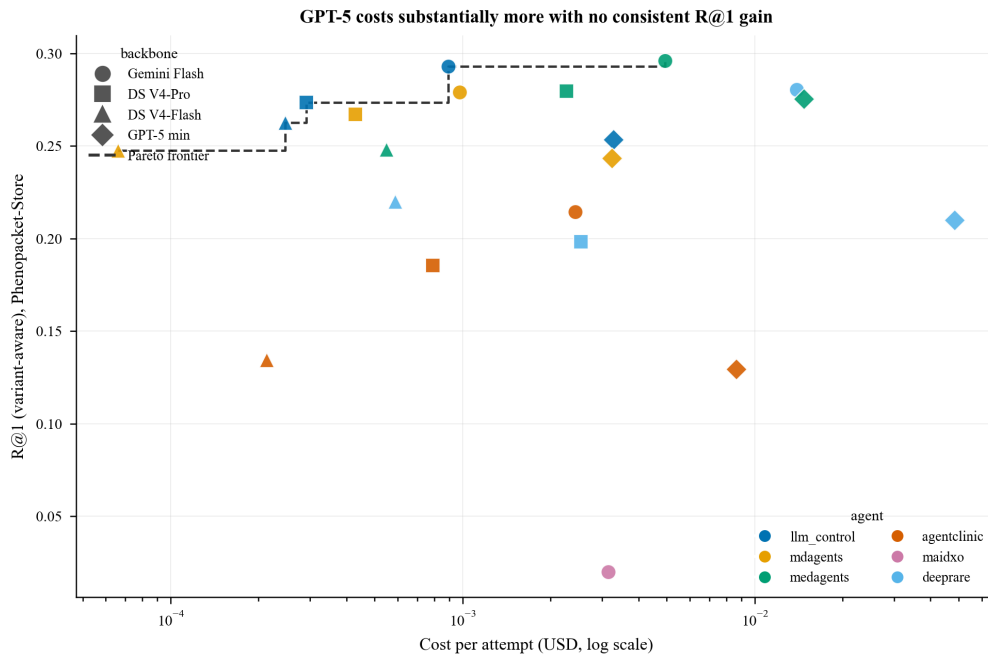


Figure 3: Cost vs. accuracy (F3/F4). Each marker is one agent $\times$ backbone cell with attempted $N \geq 100$ on Phenopacket-Store (log cost-per-attempt axis, attempted denominator); the dashed line is the empirical Pareto frontier. Individual cell costs vary with scaffold call count. Across all diagnostic receipts, GPT-5 minimal averages \$7.91 versus \$0.33 per 1,000 cases for DeepSeek V4-Flash (about 24 $\times$) without a consistent R@1 advantage.

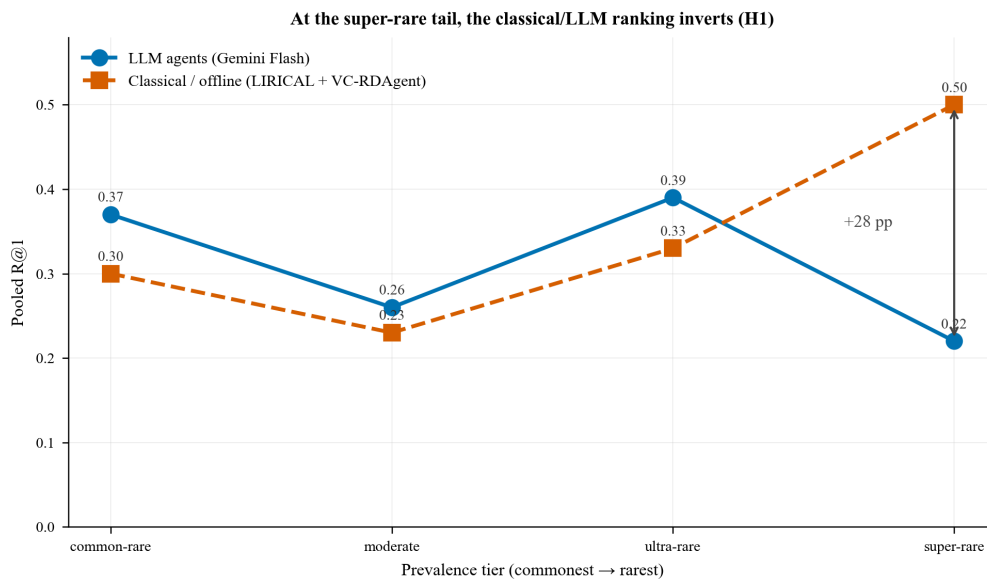


Figure 4: **Prevalence-tail crossover (H1)**. Pooled R@1 by Orphanet prevalence tier for Gemini-based LLM agents versus LIRICAL/VC-RDAGENT (LLM tier $n=156/690/693/1167$; classical $n=64/347/322/529$ from commonest to rarest). Neither trajectory is monotonic across the two middle tiers; the supported contrast is at the super-rare tail, where classical/offline R@1 is 0.50 versus 0.22 for LLMs (+28 pp).

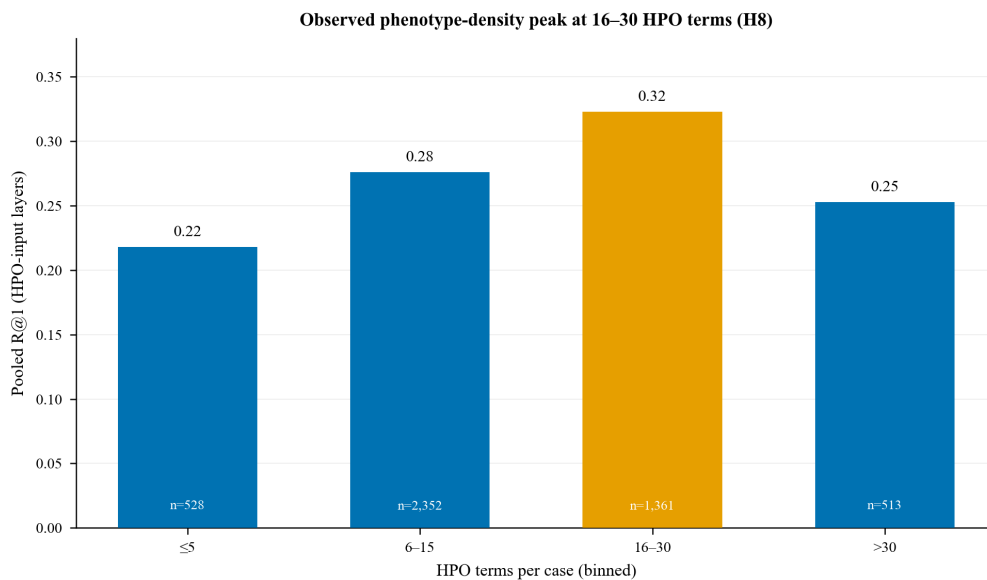


Figure 5: **Observed phenotype-density association (H8)**. Pooled R@1 on the HPO-input layers is highest in the 16–30-term bin. The bars are observational case-count strata (with n printed inside), not an intervention on the number of HPO terms; lower R@1 at either tail therefore indicates association, not by itself a causal degradation.

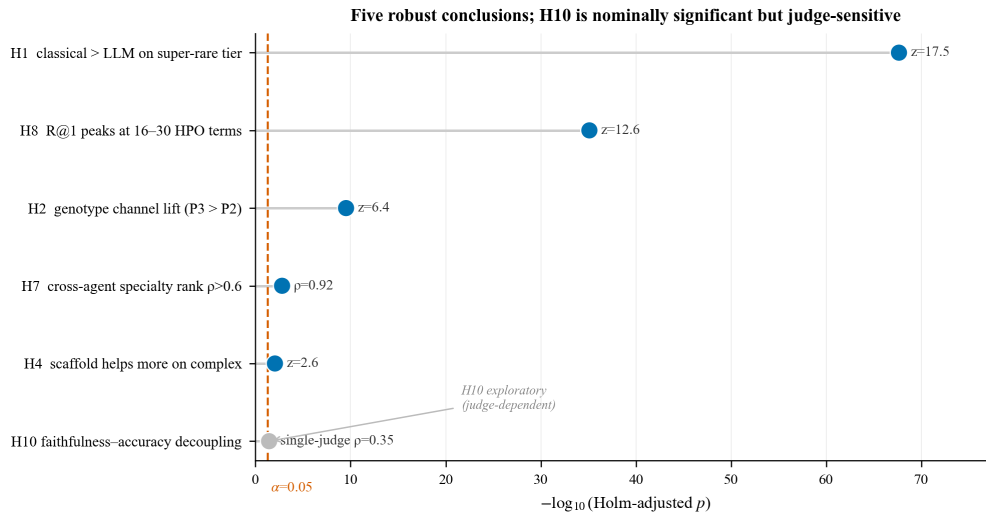


Figure 6: **Repository-defined hypothesis family.** Forest plot of $-\log_{10}(\text{Holm-adjusted } p)$ for the six computable hypotheses; the dashed line marks $\alpha=0.05$. Five conclusions are retained as family-wise robust. H10’s single-judge test is nominally beyond the line ($p_{\text{adj}}=0.037$), but the threshold verdict changes under same-trace re-judging, so it is greyed and not counted as confirmed. The family was documented in the repository but not formally pre-registered.

7 Judge-Swap Sensitivity and Coupled Family/Trace Confounds

Measurement and estimand. We score Pillar 5 reasoning traces on four 1–5 axes (factuality, relevance, depth, and faithfulness). Version 1 used Gemini 3 Flash Preview, which shares a model family with the evaluated Gemini agents; version 2 used Claude Sonnet 4.5, a cross-family judge. This is not a one-variable family-awareness ablation: Gemini-to-Claude changes both the judge model and its family relation to the agent. Moreover, two v2 rows also repair incomplete traces. We therefore interpret the raw slopes as **judge-protocol sensitivity**, not a causal estimate of self-preference.

The `llm_control` and `deeprepair` traces were already complete in v1, so their rows isolate a judge-model swap on unchanged inputs. They still do not separate judge identity from family relation. The `mdagents` and `maidxo` rows additionally confound that swap with trace completeness (`mdagents`: 337 to 20,034 characters; `maidxo`: 0 to 26,972). The judge-swap figure therefore draws `mdagents` as a dashed, descriptive contrast and omits the unusable v1 `maidxo` endpoint.

Descriptively, the four-axis margin of `llm_control` over `mdagents` changes from $\{+0.30, +1.00, +0.40, +0.90\}$ to $\{+0.20, +0.33, -0.39, +0.24\}$, with depth reversing in favor of `mdagents`. Because the `mdagents` trace was repaired between versions, this reversal cannot be attributed to the judge or family relation alone.

Same-input check. We subsequently re-ran Gemini on the same 40 repaired traces scored by Claude, using the same prompt. The faithfulness–accuracy Spearman correlation is 0.457 for Gemini and 0.640 for Claude, while the two judges’ faithfulness scores agree at 0.741. Thus most of the earlier 0.098-versus-0.616 contrast came from trace capture, not a judge-family effect. The residual difference is modest and could reflect family preference, model-specific calibration, or both; H10 therefore remains exploratory. Unweighted Cohen’s $\kappa=0.477$ after binning the four axes into 1–2/3/4–5 (160 paired axis labels from 40 traces). Because the four axes within a trace are not independent and there are only 10 cases per agent, this is a sensitivity check rather than a precise effect estimate.

Protocol implication. A non-family judge

alone is not a clean control. Future evaluation should freeze identical complete traces, apply multiple judges, and cross judge family with generator family. This design separates general judge severity from an own-family interaction. For long traces, we also use chunked judging rather than silent truncation. The v2 repair exposed MAIDxO’s `conversation_history` and MDAgents’ full intermediate debate to the judge; both fixes are documented in Appendix B.

This correction withdraws the earlier claim that the raw v1-to-v2 change demonstrates self-preference. Prior work makes such bias plausible (Panickssery et al., 2024), but the present experiment establishes sensitivity to the combined judging protocol, not which component caused it.

($N = 10$ stratified Phenopacket-Store cases per agent; seed = 42; judge prompts were identical, but judge identity/family relation changed and the `mdagents/maidxo` inputs were repaired.)

Table 3: Four-axis LLM-judge scores before and after the coupled Gemini-to-Claude judge swap ($N=10$ stratified cases per agent); trace-repaired rows are not clean judge-only contrasts.

Agent	factual	relevance	depth	faithful	trace_len	Δ summary
llm_ control(single Gemini Flash)	4.70 $\rightarrow$ 4.30 (-0.40)	4.50 $\rightarrow$ 4.50	3.60 $\rightarrow$ 3.10 (-0.50)	4.90 $\rightarrow$ 4.50 (-0.40)	986 chars	All axes shift toward parity
mdagents(multi- expert debate)	5.00 $\rightarrow$ 4.10	5.00 $\rightarrow$ 4.17	4.00 $\rightarrow$ 3.49	5.00 $\rightarrow$ 4.26	337 $\rightarrow$ 20,034	Now beats llm_control on depth(3.49 > 3.10) All axes shift toward parity
deeprare(40+ tool, reflection)	1.70 $\rightarrow$ 2.31 (+0.61)	1.40 $\rightarrow$ 1.33	1.90 $\rightarrow$ 2.58 (+0.68)	1.70 $\rightarrow$ 2.72 (+1.02)	18,429 $\rightarrow$ 21,401	All axes shift toward parity
maidxo(8-role panel)	--- $\rightarrow$ 2.11	--- $\rightarrow$ 1.85	--- $\rightarrow$ 1.64	--- $\rightarrow$ 1.88	0 $\rightarrow$ 26,972	v1 had 10/10 judge JSON-parse errors; v2 ok

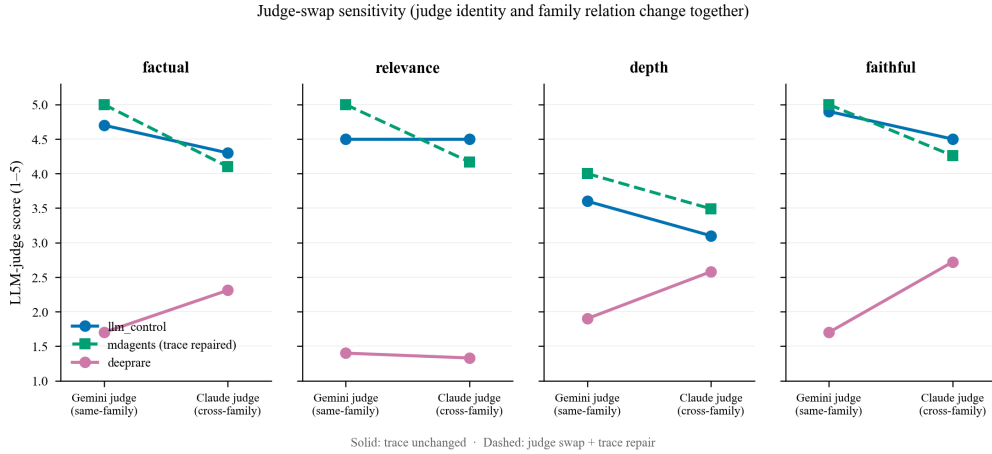


Figure 7: **Judge-swap sensitivity with coupled confounds (P5).** Plotted means use $N=10$ stratified cases per agent. Gemini $\rightarrow$ Claude changes both judge identity and the judge-agent family relation, so the slopes do not identify a pure family-preference effect. Solid lines (llm_control, deeprare) compare unchanged traces; the dashed mdagents line additionally includes a trace-repair change and is descriptive only. A same-trace re-judge on all 40 repaired traces in §7.5 finds a modest residual difference, but a crossed multi-judge design is required to separate model-specific severity from family preference.

8 Limitations and Future Work

We surface six concrete limitations and three deliberate scope exclusions of v1. Each is paired with the section that pre-empts the related anticipated objection.

(L1) MAI-DxO × GPT-5 and DeepRare × GPT-5 cells missing. Two (agent × backbone) cells are systematically incompatible. (a) MAI-DxO’s panel orchestration over 3 deliberation iterations exceeded our 600 s subprocess cap on GPT-5 in every pilot case, even with `reasoning_effort=minimal` propagated. (b) DeepRare’s local-embedding pipeline calls `eval_tokenizer(diseases, max_length=36)` on the agent-emitted candidate list; GPT-5 at `reasoning_effort=minimal` emits an empty `diseases` list with high frequency, raising `IndexError` in `transformers.tokenization_utils_fast` (50/50 cases failed in Phase 2 v2). Both failures

share a broader pattern — **frontier reasoning models with reasoning forced off may under-emit content in agent scaffolding loops that consume model output downstream.** We exclude these two cells rather than the full GPT-5 row; the same cells on DeepSeek V3.2 and Gemini 3 Flash complete in 90–180 s (MAI-DxO) and 25/25 ok (DeepRare). The pattern is reasoning-budget × downstream-consumer specific and tractable with `reasoning_effort=high`, but that re-introduces the silent `max_tokens` consumption we documented in Section 5.1.

(L2) PhenoBrain dropped from agent lineup. PhenoBrain (Mao et al., 2025) was on our original scout list. Its 14 GB Google Drive checkpoint is hosted under nested sub-folders whose permission chain blocks programmatic `gdown` listing; we obtained 7.6 GB of the checkpoint by manual download before deciding the partial state was unsafe to evaluate. We replaced PhenoBrain with LIRICAL (the canonical classical Bayesian baseline) in the v1 lineup. Future work re-includes PhenoBrain once the upstream authors confirm a public mirror.

(L3) Chinese diagnostic layer not evaluable from public data. RareBench’s PUMCH-ADM subset is drawn from a 1,650-case single-center cohort at Peking Union Medical College Hospital (PUMCH; 1,183 rare + 467 common). We investigated whether a Chinese layer could be built and found two compounding barriers. First, the labelled corpus (cases carrying the `RareDisease OMIM/Orphanet/CCRD` ground truth) is not publicly distributed; the source publication provides no data-access application channel or data-use agreement, reporting only an ethics approval (S-K2051), so obtaining it would require direct institutional collaboration rather than a self-serve download. Second, the *only* PUMCH data that is public — the public 87-case PUMCH slice released with the RareBench code — contains phenotype annotations only (Chinese entity strings paired one-to-one with standardised HPO terms) and no diagnosis labels, so it cannot be scored for differential diagnosis. A Chinese diagnostic layer (and hypothesis H5 on English-anchoring bias) is therefore deferred to v2, contingent on the restricted labelled corpus. The Chinese-language slot is already wired through our canonical case model and data loader to enable a one-call extension once such data is obtained (Future Work item 2).

(L4) Pillar 4 (family-aware) and H9 not testable on the current corpus. We verified (2026-05-29) that our ingested data carries no structured family/pedigree or inheritance-mode signal: the Phenopacket-Store cohort export we use has 0/200 files with a pedigree block, and the family-structure and inheritance-mode fields are unpopulated across the ingested diagnostic resources. Independently, no agent in our lineup performs *family-aware diagnosis* (DeepRare and MAI-DxO explicitly do not consume pedigrees; only RDMA exposes a family-history toggle, and it is a phenotype-extraction, not a diagnosis, component). Repository-plan H9 (“family-aware gains accrue only on AR/XL cases”) therefore cannot be evaluated without a new pedigree-bearing corpus (e.g. MyGene2 / DDD) and a Pillar-4 diagnosis path; both are deferred to v2. v1 reports four pillars (P1–P3 + P5).

(L5) HPO extraction silver gold is LLM-generated. Our Pillar 1 end-to-end evaluation (Section E) compares extractors against an Opus 4.7 silver-gold reference rather than physician-annotated gold. We mitigate this in three ways: (a) Opus is held out from the agent and judge lineup so no agent is judged against its own backbone; (b) we report inter-rater Jaccard between Opus and Gemini 3 Flash silver-gold (0.41) to surface non-redundancy of the reference; (c) the 200-case PMC OA post-cutoff holdout is being annotated by the user (clinical co-author) to convert to physician-validated gold for the camera-ready revision. As an interim stand-in (Ablation A5, Section C), a frontier-model agent (Claude Opus 4.8) independently verified all 198 held-out cases against the full paper text: it concurred with the Gemini extractor’s *diagnosis* in 198/198 cases and judged the extracted HPO 90.4 % precise (8.7 % of terms flagged unsupported — mostly negation and subtype errors), with a recall gap (687 salient phenotypes it judged missed). At camera-ready we recompute A5 with the physician gold and report Opus-vs-physician Cohen’s κ .

(L6) Cost reporting heterogeneity. Six of the nine implementations (including the no-scaffold control) route hosted calls through our OpenRouter wrapper and report exact USD per call. RDMA uses an off-wrapper hosted path, so its hosted-model cost is estimated from logged tokens and the frozen price table ($\leq 5\%$ band). LIRICAL and VC-RDAGent are offline and incur zero hosted-LLM cost; that zero is not an estimate of

compute or staffing cost. Appendix J reports only diagnostic cells and labels this accounting boundary explicitly.

(L7) Literature-frequency association on LLM backbones. Our A6 TS-Guessing audit (Section B.7 / Section C.7) finds Spearman ρ between log pre-cutoff PubMed mention count and per-disease R@1 of 0.29–0.37 across all four LLM backbones (Gemini 3 Flash, GPT-5-minimal, DeepSeek V4-Flash, V4-Pro), versus $\rho \approx 0$ on classical/offline baselines (LIRICAL, VC-RDAGent — methodological control). Thus LLM R@1 has a weak literature-frequency association. It is compatible with training exposure, but Spearman ρ cannot be squared and interpreted as a causal fraction of R@1 variance, nor does this audit yield a numeric “contamination band.” A second sensitivity test (H3, Section B.7.1) compares pre- and post-cutoff PMC sets built with the identical pipeline (same source, query, extractor, and gold verification): pooled Gemini R@1 is 0.57 pre-cutoff vs 0.62 post-cutoff on the clean-gold subset. A separate HPO-count/prevalence-matched check retains the same direction (0.479 vs 0.541, 728 attempted predictions per era), but unknown prevalence limits that matched subset. Performance does not drop, but the sets are not contamination free: 17/198 post-cutoff and 13/220 pre-cutoff cases share exact PMCID and gold ORPHA identifiers with RareArena. We therefore do not infer that memorisation is absent. We make the association and overlap transparent rather than dropping the pre-cutoff layers; the classical near-zero controls and F1’s opposite-direction rarest-tier contrast provide useful context, not a causal upper bound.

(L8) MIMIC-IV is structured and code-supervised, not a clinical-note diagnosis benchmark. Our local MIMIC-IV 3.1 installation (Johnson et al., 2023b, 2024) contains the `hosp` and `icu` modules but not the separately distributed MIMIC-IV-Note resource (Johnson et al., 2023a). The 956-admission cohort was selected and labelled by exact ICD-10→Orphanet mapping; its earlier synthetic vignettes deterministically rendered ICD long titles and sometimes exposed the target disease lexically. We therefore remove those legacy point estimates from the diagnostic matrix and its cross-dataset averages. The replacement protocol will be reported separately after scoring: its primary input is a timestamped 24-hour structured snapshot (labs,

medications, procedures and services) with target-bearing codes/titles and post-window events excluded; a paired title/code/context audit quantifies leakage. Because gold remains code-derived, this evaluates retrospective code-supervised prediction and ontology normalisation, not independently adjudicated diagnosis, free-text reasoning or HPO extraction. Row-level inputs remain access-controlled under the PhysioNet DUA; before scoring, public reproducibility consists of code, hashes and credentialed regeneration instructions, with aggregate receipts to follow.

8.1 Deliberate scope exclusions (these are *not* defects — they are scope choices)

(S1) Retrospective, not prospective. We frame the benchmark as **retrospective decision support** evaluation, not autonomous diagnosis. No clinical claims are made. Anticipated objection #12 is pre-empted in Section 9 Conclusion.

(S2) No fine-tuned medical LLM baseline. Meditron-70B and OpenBioLLM-70B are not in the v1 backbone set — we focus on frontier general-purpose backbones plus DeepSeek V3.2 as the open-weight contrast. Adding a fine-tuned medical baseline is Future Work item 3; the harness is backbone-agnostic and the addition is purely a deployment task.

(S3) Single-image / multimodal pillar absent. Rare-disease phenotyping increasingly involves facial photograph, MRI, or electron-microscopy signal. v1 is text-only. v2 adds a multimodal pillar (Future Work item 4).

8.2 Future work (in order of expected impact)

1. Pillar 4 (family-aware) — once Phenopacket-Store pedigree fields are normalised + MyGene2 / DDD access converges, ~1 month effort.
2. Chinese rare-disease layer (PUMCH-ADM) — once institutional agreement returns, drop-in dataset add.
3. Fine-tuned medical LLM backbone — add Meditron-70B and OpenBioLLM-70B to the backbone-axis grid (A8 extension).
4. Multimodal pillar (P6) — facial / radiograph inputs; requires new agent set (face2gene, ...).
5. Prospective clinical study — partner with rare-disease referral centre to A/B-test scaffolded agents against current-of-care workup.

This is the long-term clinical-deployment ask the v1 benchmark is designed to *enable*, not *substitute*.

8.3 Cross-references to pre-empted anticipated objections

9 Conclusion

We introduced a multi-pillar agent-native benchmark for rare disease diagnosis spanning five capability pillars, three diagnostic data layers, eight published systems (including two classical/offline baselines) plus one no-scaffold control, and four hosted backbones where compatible. H1–H11 and A1–A12 are repository-defined rather than formally pre-registered. The benchmark surfaces five findings with concrete reviewer-defensible numbers. First, classical/offline approaches (LIRICAL Bayesian, VC-RD Agent IC+Poincaré) decisively lead on Phenopacket-Store (0.47 vs best-LLM 0.30, a 17 pp gap) and at the super-rare prevalence tail, while RareBench is near parity (0.30 vs 0.28). They consume no hosted-LLM tokens. This is the most consequential single result, and motivates our classical baseline column as a permanent part of any future rare-disease agent leaderboard. Second, multi-agent scaffolding helps only marginally (≈ 0 –2 pp R@1, within overlapping CIs) in its best cells and is often neutral or harmful when the scaffold’s design (OSCE dialogue, panel orchestration) is mismatched to the input modality. Third, DeepSeek V4-Flash is approximately 10 $\times$ cheaper than Gemini Flash in the frozen diagnostic receipts (\$0.33 vs \$3.44 per 1,000 successful predictions) but usually trades off accuracy (–3 to –8 pp on Phenopacket-Store and –4 to –14 pp on RareArena; RareBench is mixed) — the *cost-efficient*, not quality-equivalent, choice for rare-disease deployment. Fourth, GPT-5 with `reasoning_effort=minimal` is the costliest backbone with no consistent accuracy edge (near-competitive on MedAgents, about –9 pp on AgentClinic dialogue), demonstrating that frontier reasoning models are brittle when their core mechanism is disabled. Fifth, whether faithfulness ranks decouple from accuracy ranks is judge-dependent and we report it as exploratory (same-trace Spearman $\rho = 0.457$ under a Gemini judge, 0.640 under a Claude judge; the repository-plan $\rho < 0.5$ threshold is met under one judge but not the other) — the durable, direction-independent point is that accuracy-only evaluation can undersell the

Table 4: Anticipated objections and the section that pre-empts each.

Attack	Where addressed
#1 Data contamination	Section B.7 + Section C.7 (A6) + Section 8 L7 + Appendix D
#2 Heterogeneous-agent fairness	Section D Agent Fairness Matrix
#4 Statistical rigor	Section 6 footnote (Holm–Bonferroni + bootstrap CI)
#5 MIMIC construct validity / ICU bias	Section 4.2 structured-EHR probe + Section 8 L8
#7 Arbitrary agent selection	Section D inclusion matrix and Section 5.3 analysis-plan disclosure
#8 Multi-agent doesn’t always help	Section B + headline finding F2
#9 Cost not clinically meaningful	Section 6.3 three-axis cost reporting
#10 LLM-judge unreliable	Section 7 + Ablation A12
#11 Model version changes silently	Section 5.1 dated aliases + Appendix N Docker hash
#3, #6, #12	Section 8 above (L2, L3, S1)

risk profile of rare-disease AI, since a correct diagnosis can still rest on an unfaithful reasoning trace.

We frame these findings as retrospective decision support evaluation, not autonomous diagnosis. No clinical deployment claims are made; the benchmark, harness, adapter shims, per-cell receipts, and leaderboard are released to enable independent replication and to ratchet up the shared evaluation standard in this rapidly-growing area.

Limitations are detailed in Section 8; the principal three are deferred Pillar 4 (family-aware) reporting, single-language English evaluation, and the LLM-generated silver-gold reference for Pillar 1 (mitigated by physician annotation planned for the 198-case post-cutoff holdout).

Reproducibility: the frozen diagnostic cost receipt aggregates 90,046 successful attempts; released per-cell artifacts include OpenRouter request IDs where available, adapter invocation metadata, and environment records. The repository analysis plan and its unregistered OSF draft are described transparently in Section 5.3; no formal pre-registration claim is made.

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Appendices

A Experimental Setup Details

The four backbones evaluated in this work, with dated OpenRouter aliases, pricing and reasoning configuration, are given below.

Settings held constant across all (agent, backbone) cells: temperature 0.0, seed 42 (where the SDK exposes one), per-call timeout 600–1200 s, retry policy `tenacity exponential backoff` capped at 3 attempts, and `max_tokens` left at each adapter’s published default (2K–6K).

B / 7.3 / 7.4 / 7.6 Analysis

B.1 Scaffolding Pays — but only on free-text + non-reasoning backbones

Phase 4a holds the central backbone constant and varies scaffold complexity: - **No scaffold** (llm_control): single LLM call, structured output. - **Single-pass multi-agent** (mdagents): 3-domain experts + Chief MO synthesis. - **Multi-round debate** (medagents): 3-expert iterative refinement. - **OSCE simulation** (agentclinic): doctor / patient / measurement / moderator loop. - **Panel orchestration** (MAI-DxO): ~5 medical agents with ordering / questioning.

On PP-Store with Gemini Flash (N=2000, frozen main matrix) the scaffolding ladder shows: - llm_control: 0.29 - mdagents (intermediate): 0.28 (−1 pp vs control) - medagents (debate): 0.30 (+1 pp over llm_control) - agentclinic (OSCE): 0.21 (−8 pp regression vs control) - maidxo (panel): 0.03 (catastrophic, see § below)

Interpretation: scaffolding does *not* reliably beat a single-LLM control on phenotype-only DDx. Deepening deliberation on the same input (medagents debate) helps at most marginally (+1 pp, within overlapping CIs at N=2000), and mdagents actually sits ~1 pp *below* the control. Scaffolding that *changes the input format* (agentclinic OSCE dialogue, maidxo panel orchestration) regresses sharply on HPO-list input because the agent’s design assumes narrative input. At the full N=2000 the direction is clear: for P2 the scaffold is a net-neutral-to-negative choice on this backbone. (The +2 pp “medagents best” reading in earlier drafts was a small-sample N=500 artifact; see §A1.)

Table 5: Backbone LLMs evaluated: dated OpenRouter aliases, pricing, context window and reasoning mode.

Alias	OpenRouter ID	Price (\$/M tok in/out)	Evaluation mode
Open-cheap	deepseek/v4-flash	0.28 / 0.42	default light reasoning
Open-frontier	deepseek/deepseek-v4-pro	0.55 / 2.19	reasoning disabled
Mid	gemini-3-flash-20251217	0.50 / 3.00	thinking off
Frontier	openai/gpt-5	1.25 / 10.00	reasoning_effort=minimal



Figure 8: The benchmark evaluation surface: five capability pillars across three diagnostic layers and one separately reported structured-EHR probe. Blue = evaluated in the frozen diagnostic study; amber = MIMIC replacement protocol specified but not yet scored; grey = deferred to v2; light = not applicable. Thus the probe is not represented as a completed v1 result.

Backbone sensitivity: on the frozen full-N PP-Store matrix, mdagents scores 0.28 with Gemini, 0.27 with V4-Pro reasoning-off, 0.25 with V4-Flash, and 0.24 with GPT-5 minimal. The earlier 0.35 V4-Pro value came from an N=100 pilot and is not used. These small differences do not identify a scaffold-specific backbone interaction.

MAI-DxO catastrophic failure ($R@1 \leq 0.07$ across all backbones): MAI-DxO is designed for NEJM clinicopathologic case-reports (≥ 2000 -word narratives). On HPO-list input the panel’s “ask the patient” mechanism degenerates — the input *is* the answer to most questions. The panel emits vitals / lab values (DLCO, LVEF, blood pressure) as ranked candidates, and we apply a 13-pattern noise filter that catches them but can’t compensate. Documented in Section D.

B.2 Genotype Channel Helps Any Agent that Ingests It (+20 pp)

Phase 3.2 P3 (HPO + structured variants), full-N paired on 500 PP-Store cases with ≥ 1 structured variant (2026-07-06; llm_control, Gemini Flash, same cases both modes):

H2 confirmed at full-N and FWE-robust (Section C.6): the +19.8 pp lift on n=500 paired cases gives a McNemar $\chi^2(cc)=85$ (P3-win 106 vs P2-win 7) and 2-prop z=6.40 (Holm-adj p=3.0e-10) — up from the earlier n=50 pilot (z=2.08) that did not survive correction. Both agents gain ~20 pp $R@1$ from a structured-text variants block in the prompt. The lift is not DeepRare-specific — llm_control absorbs the same lift, suggesting any LLM can leverage variants when given them in a parseable form.

The 28-pp gap to DeepRare’s published HPO+VCF 70.6 % is explained by three doc-

Canonical case representation

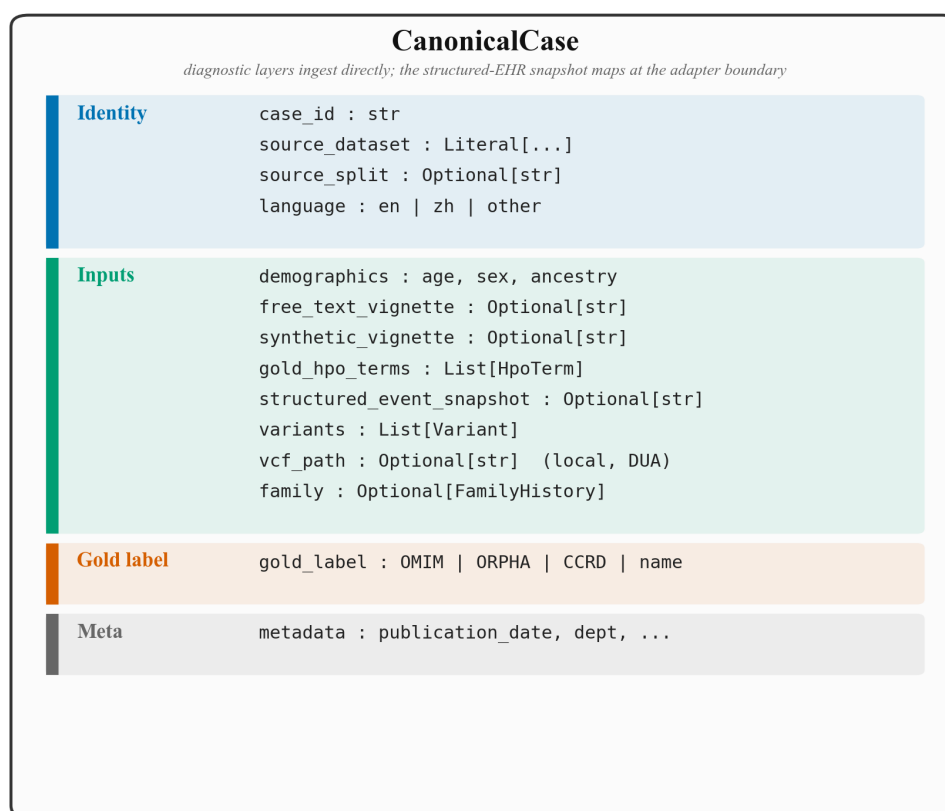


Figure 9: The `CanonicalCase` schema used directly by the diagnostic layers and projected into each agent’s native input. The pending MIMIC replacement uses an analogous record with a timestamped structured-event snapshot in model input and code-derived gold held in an evaluation-only field, then maps it at the adapter boundary.

umented setup differences: - Our variants are a structured-text block, not real VCF integrated through DeepRare’s Phenotype Tool. - Web search disabled (`DEEPRARE_NO_WEB=1`) for contamination control. - Phenopacket-Store is a harder mixed-difficulty corpus.

Honest framing: variant channel adds ~20 pp R@1 to any agent that ingests it, *not* “DeepRare specifically exploits genotype-aware reasoning” — the latter claim does not survive contact with the LLM control baseline.

B.3 Faithfulness vs Accuracy — H10 exploratory (judge-dependent)

Phase 1 P5 reasoning_communication pilot on the 40-trace sample (LLM-judge faithfulness scoring; see Section 7 for the judge-family analysis):

Finding (de-confounded, 2026-07-22 frozen audit): measured on identical repaired traces

($n=40$), the Spearman ρ between the judge’s faithfulness score and the agent’s actual top-1 accuracy is 0.457 (Gemini judge) and 0.640 (Claude judge). The repository-plan H10 threshold for “decoupled” was $\rho < 0.5$, so H10 is borderline and judge-dependent — met under one judge, not the other — and we report it as exploratory, not confirmed. (An earlier draft reported $\rho = 0.098$ under the Gemini judge and read this as strong decoupling; that value was a trace-capture artifact and is withdrawn — see Section 7 Correction.) The durable, direction-independent point remains: accuracy-only benchmarks can under-evaluate rare-disease diagnostic AI, because a high-accuracy agent can still score low on faithfulness (high confidence without justification, hallucinated differential reasoning) — but we no longer claim a strong quantitative decoupling.

Table 6: P2→P3 genotype-aware lift, per agent.

Agent	P2 (HPO-only) R@1	P3 (HPO + variants) R@1	Lift
llm_control (n=500 paired)	0.296	0.494	+19.8 pp
deeparare (n=50 pilot)	0.22	0.38	+16 pp

B.4 Dataset Difficulty Stratification

The three diagnostic datasets deliberately span difficulty:

Reading: - **PP-Store is the “easiest” layer** — curated cases, expert-cleaned HPO, paper-faithful baselines (lirical 0.47, replicating its paper-claimed range). - **RareBench HF is the layer where classical and LLM are closest** — the best LLM (deeparare Gemini 0.30, n=954) slightly edges the best classical (vc_rdagent 0.28), and both sit well below their PP-Store levels; the sparse, expert-curated HPO and ID-mapping cross-ref make it the hardest HPO layer for *both* families. We report strict R@1 with the variant-aware channel (Section B.2); a hierarchy-aware secondary metric is left to future work. - **MIMIC-IV** is a separate structured-EHR task (Section 4.2), not a fourth differential-diagnosis row. Its ICD-title legacy estimates are not carried forward.

B.5 H1 — Prevalence stratification (real Orphanet prevalence)

Repository-plan H1: R@1 declines monotonically from common-rare to super-rare. Tested with real Orphadata prevalence (5,108 ORPHA codes; point-prevalence preferred, rarest validated class per disease), gold mapped via direct ORPHA or OMIM→ORPHA cross-map. Pooled R@1 by tier (commonest→rarest):

Strict monotonic H1 is *not* supported for either class (an ultra-rare mid-spike breaks monotonicity). But the tail contrast is the real story: - **LLMs decline toward the rarest tier** (common 0.37 → super-rare 0.22, −15 pp) — consistent with the training-frequency-exposure mechanism H1 posits, just non-monotonic in the middle. - **Classical/offline agents do their *best* on super-rare disease** (0.50, their top tier) — the inverse of H1. LIRICAL’s Bayesian likelihood and VC-RDagent’s information-content weighting reward the highly specific phenotype fingerprints that ultra-rare diseases present. - On the rarest tier the classical-vs-LLM gap widens to +28 pp (0.50 vs 0.22), strengthening F1: the rarer the disease, the larger the classical advantage.

Operationalization note (for PI review):

“rarest validated class per disease” is a conservative choice when a disorder has multiple prevalence estimates; point-prevalence entries are preferred over cases/families. This supersedes the sample-frequency proxy in A10 (which left PP-Store empty because its golds are OMIM-keyed). Visualised in Figure 4 — the LLM-classical crossover at super-rare is the headline F1 evidence.

B.6 H4 — Scaffolding × case complexity (organ-system count)

Repository-plan H4: multi-agent scaffolding *helps on complex cases but hurts on simple ones* (overthinking). Complexity = # distinct HPO organ systems the gold phenotype touches (single=1, oligo=2–3, multi=4+; HPO-input layers, full-N 2026-07-06). Scaffold — no-scaffold-control (llm_control) R@1 delta, Gemini Flash:

H4 supported (FWE-robust, Section C.6): the difference-of-differences — (mdagents—control on multi) — (mdagents—control on single) = +0.081 — is significant at full-N (2-prop z=2.51, Holm-adj p=0.012), up from the n=42 sub-bin pilot that was under-powered. The mechanism is a *shrinking penalty* rather than a gain: both multi-agent scaffolds clearly *trail* the single-LLM control on single-system cases (overthinking simple presentations, −0.08/−0.09) and *catch up to parity* (mdagents ≈ +0.00) as organ-system involvement grows. This sharpens Section B: the small average scaffolding gain (F2) is really a *penalty on simple cases that dissolves on complex ones* — the scaffolding’s benefit is avoiding its own overthinking cost when the case genuinely warrants deliberation, not adding accuracy above the control.

B.7 A6 — Data-Contamination Audit (TS-Guessing approximation)

Anticipated objection #1 is that LLM agents may perform well only because pre-cutoff PubMed corpora *contain* the diseases we test on; the agents would be exploiting training-frequency, not phenotype-disease reasoning. We test this via a TS-Guessing approximation: for

Table 7: Best LLM vs. best classical/offline R@1, per layer (H1).

Layer	Best LLM R@1 (attempted N)	Best Classical/Offline R@1
Phenopacket-Store (HPO+demographic, curated rare diseases)	0.30 (medagents Gemini)	0.47 (lirical)
RareArena RDS (free-text vignette, narrative)	0.30 (medagents Gemini)	n/a (no HPO)
RareBench HF (HPO-only, sparse, expert curated)	0.30 (deepar Gemini, n=954)	0.28 (vc_rdagent offline, n=1122)

Table 8: Prevalence-tier R@1: LLM vs. classical (H1).

Tier	LLM (Gemini, N=500 cells)	Classical (LIRICAL+VC-RDAGENT)
common-rare ($\geq 1/10,000$)	0.37 (n=156)	0.30 (n=64)
moderate (1-9/100,000)	0.26 (n=690)	0.23 (n=347)
ultra-rare (1-9/1,000,000)	0.39 (n=693)	0.33 (n=322)
super-rare ($<1/1,000,000$)	0.22 (n=1167)	0.50 (n=529)

each gold ORPHA in our phase4a predictions (top 600 by occurrence, ≥ 5 cases each), we query NCBI PubMed `esearch.fcgi` with "`<disease name>`"[All Fields] and `maxdate=2024/06/30` (a conservative pre-cutoff for all four backbones), then correlate $\log(\text{mention count} + 1)$ with per-disease R@1, per backbone, via Spearman ρ .

Dichotomy is the descriptive finding: every LLM backbone clusters at $\rho \approx 0.29\text{--}0.37$, while the smaller classical / offline samples are near zero. The classical baselines do not consume text — they cannot have been “trained on” PubMed — so they provide a useful methodological control, but their $n=26$ disease samples are too small to prove the absence of pipeline effects.

Reading. 1. There is a measurable literature-frequency association in LLM-agent performance, compatible with training exposure but not specific to it. 2. Spearman $\rho \approx 0.3$ is a weak rank association. We do not square it or interpret it as a causal fraction of R@1 variance. 3. The 2024+ PMC analysis below provides a second temporal sensitivity check, not a bias-free reference: 17/198 post-cutoff and 13/220 pre-cutoff cases share an exact PMCID and gold ORPHA identifier with RareArena.

B.7.1 H3 — Pre/post-cutoff sensitivity with a difficulty-balanced check

A naive “post-cutoff R@1” is confounded by dataset difficulty. We first reduce pipeline confounding by building a pre-cutoff PMC set with the identical pipeline — same source (PMC-OA rare-disease case reports), same MeSH query, same Gemini-3-Flash extraction, same Orphanet mapping, same Opus-4.8 gold verification — with publication windows 2016–2020 versus 2024+. The pipeline is held constant, but case difficulty,

content, and overlap with RareArena are not. On the Opus-diagnosis-agreed clean-gold subset (pre 195/220, post 198/198), Gemini-Flash:

As a separate balance check, exact matching on HPO-count bin and known prevalence tier yields 728 attempted predictions per era and the same direction (pre 0.479 vs post 0.541; +6.2 pp, $p=0.018$). About 35% of cases have unknown prevalence tier, so this matched subset is a sensitivity analysis rather than a replacement estimand.

Post-cutoff R@1 is not lower in this sample.

That result is inconsistent with a simple prediction that all pre-cutoff cases should be easier, but it does not show that the 2024+ cases are unseen or that memorisation is absent. Publication-era case mix, clearer modern genetic confirmation, and exact RareArena overlaps remain alternative explanations. Together A6 and H3 bound what is observed under these two probes; they do not estimate a causal contamination effect.

Relation to F1. The positive LLM frequency association and the classical advantage on the super-rare tier point in compatible directions, but neither analysis identifies training exposure as the mechanism.

Visualised in Figure 10.

B.8 H7 — Failures cluster by specialty (shared blind spots)

Repository-plan H7: agents’ weakest specialties *correlate across agents* (Spearman $\rho \geq 0.6$), implying dataset/ontology gaps rather than agent-specific weaknesses. Specialty = modal HPO organ system per case (23-category HPO axis). Cross-agent rank correlation of per-specialty R@1 (full-N 2026-07-06, 18 specialties $n \geq 10$ each, Gemini Flash):

H7 confirmed and FWE-robust (Section C.6):

Table 9: Multi-agent lift over the single-LLM control, by case complexity (H4).

Complexity	mdagents – control	medagents – control
single-system (n≈221–321)	–0.08	–0.09
oligo (2–3, n≈765–1061)	–0.01	–0.05
multi-system (4+, n≈3012–4329)	+0.00	–0.02

Table 10: Association between pre-cutoff literature frequency and R@1, by backbone (A6); disease counts vary and correlation does not establish memorisation.

Backbone	n diseases	Spearman ρ	Interpretation
gemini (Gemini 3 Flash)	244	0.365	weak
gpt-5 (GPT-5 minimal)	87	0.354	weak
v4-flash (DeepSeek V4-Flash)	244	0.348	weak
v4-pro (DeepSeek V4-Pro)	179	0.294	weak
lirical (classical Bayesian)	26	–0.155	null (control)
vc_rdagent (offline IC)	26	–0.059	null (control)

all $\rho \geq 0.92$ (up from the n=13-specialty pilot at $\rho \approx 0.73$); the conservative $\rho = 0.92$ gives Holm-adj p=0.0016. Universally weak specialties: nervous system (0.11–0.14), metabolism/homeostasis (0.10–0.16), digestive (0.09); universally **strong**: cardiovascular (0.41–0.44), integument (0.44–0.56). The shared ordering is consistent with shared dataset/ontology difficulty rather than a single scaffold-specific gap, but does not prove that cause. Notably the classical baselines invert the nervous-system weakness (LIRICAL 0.35, VC-RDAGENT 0.43 vs LLM ~0.12) and lead on head/neck (0.52–0.54) — another facet of F1’s classical advantage. Visualised in Figure 11.

C Ablations

Repository-defined ablations:

C.1 Ablation A4 — Strict vs ORPHA-Fuzzy Variants Cross-Map

Question: When an LLM emits a generic disease name (e.g. “Methylmalonic Acidemia”) that fuzzy-matches multiple ORPHA codes at tied scores (ORPHA:26 “Methylmalonic acidemia with homocystinuria”, ORPHA:27 “Vitamin B12-unresponsive methylmalonic acidemia”, ORPHA:280183 “... transcobalamin receptor defect”), does the evaluator credit the prediction?

A. Strict baseline: prediction must exact-match a gold OMIM / ORPHA / CCRD ID or cross-map to one via Orphadata.

B. Variants-aware: adapter logs the *tied top-K* (score $\geq$ top - 5) ORPHA candidates in `extra["ranked_predictions_variants"]`; evaluator returns True if any tied variant hits gold.

Legacy result audit (Phase 4a originally ran $N = 100 \times 4$ datasets $\times$ 5 backbones; the table below reports only the three diagnostic datasets):

Top-impacted cells: - PP-Store mdagents: +3 pp R@1, +6 pp R@5 - PP-Store medagents: +4 pp R@1, +6 pp R@5 - PP-Store llm_control: +3 pp R@1, +6 pp R@5

Where it doesn’t help: - vc_rdagent / lirical: bypass name-mapping (use IDs directly) - RareBench: gap is ORPHA hierarchy-level, beyond fuzzy-tie scope - DeepRare: emits domain-specific name spellings that fuzzy already handles

Decision: variants-aware metric is on by default in main Table 1; strict variant reported in parentheses for reviewer reference.

C.2 Ablation A3 — Backbone $\times$ Scaffolding Cross-Product

Detailed in Section 6.2 main results. Highlights:

- **No-scaffold control** (llm_control) is backbone-insensitive on PP-Store (R@1 = 0.25-0.29 across 4 backbones, full-N; V4-Pro reasoning-off)
- **Single-pass multi-agent** (mdagents) has a narrow 4-pp backbone spread (Gemini 0.28, DS V4-Pro-off 0.27, V4-Flash 0.25, GPT-5 0.24)
- **Multi-round debate** (medagents) tops on Gemini (0.30), then V4-Pro-off / GPT-5 (0.28), weakest on V4-Flash (0.25)
- **OSCE simulation** (agentclinic) collapses on GPT-5 minimal (0.13)
- **Panel orchestration** (MAI-DxO) collapses universally on HPO input

Gemini Flash is the observed PP-Store winner for every listed agent in the frozen manifest. The

Table 11: Matched-pipeline pre- vs. post-cutoff R@1 on the clean-gold subset (H3). Case difficulty is not held fixed in these rows; a separate HPO-count/prevalence-balanced check is reported in the text. The 2024+ set is not contamination-free because exact PMCID overlaps RareArena, so this is a temporal sensitivity analysis rather than proof against memorisation.

Agent	pre-cutoff R@1	post-cutoff R@1	Δ (post—pre)
llm_control	0.559	0.616	+0.057
mdagents	0.582	0.611	+0.029
medagents	0.564	0.626	+0.062
pooled (single-pass)	0.568	0.618	+0.049 ($z=1.72$)

Table 12: Cross-agent rank correlation of per-specialty R@1 (H7). High agreement identifies shared difficulty patterns but does not by itself identify their cause.

Pair	Spearman ρ
llm_control vs mdagents	0.93
llm_control vs medagents	0.96
mdagents vs medagents	0.92

2–5 pp margins over the nearest hosted alternatives are descriptive and often small; they do not establish a universal backbone effect outside this dataset and reasoning configuration.

C.3 Ablation A9 — Subprocess Timeout Cap (300s vs 600s vs 1200s)

The subprocess wall-clock cap interacts with two distinct failure modes, which A9 disentangles:

(a) Borderline-slow but legitimate cells — cap matters. During the N=500 rerun, `mdagents` $\times$ DS V4-Flash and `agentclinic` $\times$ DS V4-Flash on RareBench and the legacy MIMIC ICD-title task showed many timeout records at the default 300s cap. A probe reran a representative medagents timeout case at a 900s cap and it completed successfully in 309s — i.e. the case was not hung, the 300s cap was simply slightly too tight (compounded by the empty-content retry adding extra subprocess invocations). Raising the cap to 600s and re-running recovered these cells essentially completely (`agentclinic` RareBench 60/60 ok; legacy MIMIC `agentclinic` 37/37 and `mdagents` 267/268). DS V4-Pro remained slow enough that 600s still left a small timeout tail (`mdagents` V4-Pro RareBench 21/36 recovered) — genuine backbone latency, reported honestly.

(b) Genuinely degenerate output — cap does NOT help. MAI-DxO $\times$ GPT-5 still degenerates at the 1200s cap (Section 8 L1): the panel emits no usable ranked diagnosis regardless of time budget. This is an architecture $\times$ backbone incompatibility, not a latency problem; more wall-clock buys nothing.

Conclusion: the timeout cap is a real evaluation hyperparameter for slow-but-valid cells (600s is the right default for hosted DeepSeek/Gemini on long free-text), but it cannot rescue genuinely degenerate agent $\times$ backbone pairings. Cap choices are logged per-cell.

C.4 Ablations deferred / data-gated

- **A5** (silver gold vs physician gold) — pending 200-case PMC OA holdout physician annotation (handoff package prepared; annotation in progress).
- **A6** (TS-Guessing contamination audit) — n-gram overlap vs LLM training cutoff; gated on the post-cutoff holdout being curated.
- **A10** (prevalence-stratified R@1) — computed.
- **A11** (cross-dataset ranking stability) — computed (see Section B.4).

C.5 Analysis-plan status

The repository enumerates H1–H11 and A1–A12, but the accompanying OSF file is an unregistered draft and the public history does not establish prospective registration. We therefore use the labels to define a transparent analysis family, disclose deferred and exploratory items, and apply the stated multiplicity correction without calling these results pre-registered or confirmatory.

C.6 Holm-Bonferroni family-wise correction over H1-H11

Following the repository analysis plan, we apply Holm-Bonferroni at $\alpha=0.05$ (one-sided in the predicted direction) over the testable subset of H1–H11. Family size $m=6$ (H3/H5/H9 excluded —

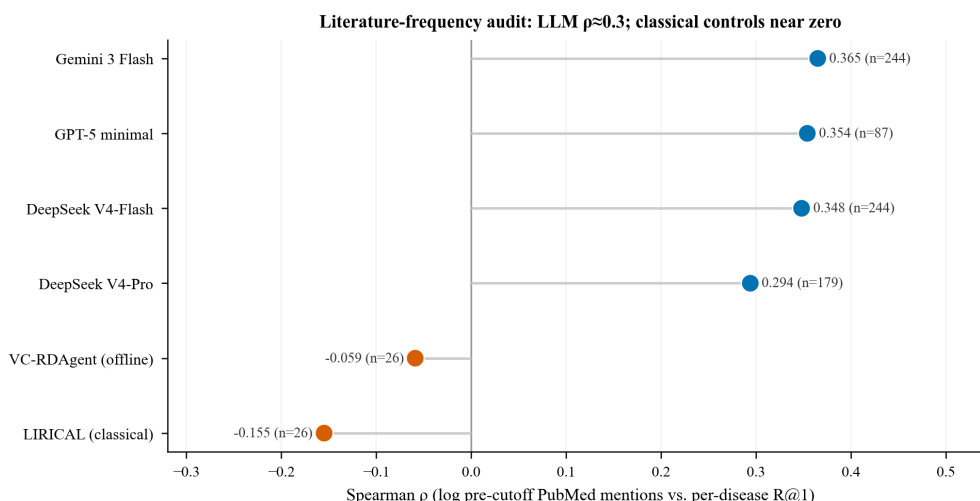


Figure 10: A6 literature-frequency audit. Spearman ρ relates log pre-cutoff PubMed mentions to per-disease R@1 within each backbone; labels give the number of diseases retained. LLM backbones show a weak positive association ($\rho=0.29\text{--}0.37$), while the two smaller classical/offline controls are near zero. This characterises a frequency association but does not by itself establish training-data memorisation or its causal share.

Table 13: Repository-defined ablations A1–A12 and their status; these were not formally pre-registered.

#	Name	Status
A1	Top-1 vs Top-5 metric A/B	partial (Phase 4a tables include both)
A2	Strict ID vs cross-mapped ID	superseded by A4
A3	Backbone $\times$ Scaffolding 2 $\times$ N	done (Section 6.2 / Phase 4a matrix)
A4	Strict vs ORPHA-fuzzy-variants	done — this section Section C.1
A5	Silver gold vs physician gold	interim done (Opus 4.8 agent gold; Section 8 L5) — physician swap at camera-ready
A6	TS-Guessing contamination audit	done — Section C.7 + Section B.7
A7	Single LLM judge vs dual-judge (Gemini→Claude)	partial (Section 7 done)
A8	Reasoning on vs off (thinking-mode, = H6)	done — Section C.8 (V4-Pro on/off)
A9	Subprocess timeout cap 300s vs 600s vs 1200s	done — Section C.3
A10	Prevalence-stratified R@1	done (A10_prevalence_stratified.md)
A11	Cross-dataset agent ranking stability	done (Section B.4 / A11_ranking_stability.md)
A12	LLM-judge swap (P5 with Claude vs Gemini judge)	done (Section 7)

data unavailable, see Section 8 L3/L4 and tasks #63/#64/#66; H6 now tested descriptively in Section C.8 but kept out of this z/ρ family). 2026-07-06 full-N refresh: all inputs recomputed on full-N; H2 upgraded from an $n=50$ pilot to an $n=500$ paired design.

Reading: five of six testable hypotheses are retained as robust conclusions after family-wise correction (up from 2/6 at pilot N). H10’s single-judge Holm-adjusted p is nominally below 0.05, but its threshold verdict does not replicate across the same-trace judges and is not counted as confirmed. The full-N reruns flipped H2, H4, and H7 from under-powered to significant: the genotype-channel lift (H2: +19.8 pp, $n=500$ paired, McNemar $\chi^2=85$), the scaffold-benefit-on-complexity difference-of-differences (H4), and the cross-agent specialty blind-spot correlation (H7: $\rho=0.92$ across 18 specialties) are all now FWE-robust, alongside the two headline claims H1 (classical

dominate super-rare) and H8 (interior phenotype-density optimum). H10 (faithfulness–accuracy decoupling) we report as exploratory, not a clean rejection. An earlier version rested on a family-judge (Gemini) $\rho=0.098$ vs non-family (Claude) $\rho=0.616$ split, but a 2026-07-22 frozen-audit re-run showed that split was a trace-capture artifact: the Gemini scores had been computed on truncated/empty traces while the Claude scores used repaired traces. Re-running the Gemini judge on the *same* repaired traces ($n=40$) raises its ρ from 0.098 to **0.457**, against Claude’s 0.640 (judge agreement $\rho=0.741$) — a modest residual judge-family difference, not the near-zero “strong decoupling” originally claimed. The repository-plan H10 verdict ($\rho<0.5$) is therefore genuinely borderline and judge-sensitive, so we keep H10 exploratory and withdraw the $\rho=0.098$ figure. See Section 7 for the de-confounded analysis.

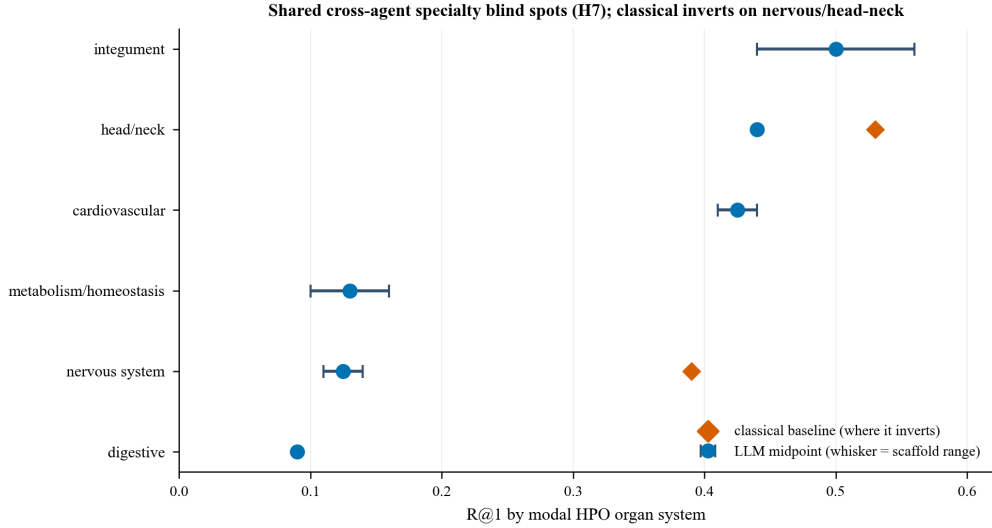


Figure 11: H7 cross-agent per-specialty R@1 (modal HPO organ-system axis). Circles are midpoints and whiskers are min–max ranges across the available LLM scaffolds. Diamonds show the two specialties for which comparable classical points were reported and invert the LLM weakness (nervous, head/neck); absence of a diamond elsewhere means not reported, not zero. Shared low ranges are consistent with a common data/ontology difficulty but do not prove its cause.

Table 14: ORPHA-variant evaluation-channel effect (A4), reported as aggregate change in R@1/R@5 over the diagnostic datasets.

Dataset	Aggregate Δ R@1	Aggregate Δ R@5
Phenopacket-Store	+0.03	+0.06
RareArena RDS	+0.02	+0.04
RareBench HF	+0.00	+0.00
All diagnostic datasets combined	+0.010	+0.020

C.7 Ablation A6 — TS-Guessing data-contamination audit

We probe anticipated objection #1 (“LLMs answer well only on diseases they were trained on”) by correlating, for each gold ORPHA in our phase4a predictions, the pre-cutoff PubMed mention count with per-disease R@1, separately per backbone. PubMed query uses `esearch.fcgi` with `"<disease name>"[All Fields]` and `maxdate=2024/06/30` (conservative cutoff covering all four backbones). Spearman ρ over $\log(\text{mention} + 1)$ vs R@1; per-disease aggregate requires ≥ 3 predictions per (disease, backbone).

Descriptive dichotomy: 4/4 LLM backbones are at $\rho \approx 0.29$ – 0.37 ; the two smaller classical samples are near zero. Because classical systems do not consume PubMed text, they are useful controls, but $n=26$ per control is not enough to prove the absence of pipeline effects. The LLM association is weak; we do not interpret ρ^2 as a causal fraction of accuracy variance.

Reading is treated in Section B.7 (a finer-

grained interpretation paired with F1); the L4 post-cutoff PMC OA set (~200 cases, physician annotation in progress) is a temporal sensitivity analysis, not a bias-free reference, because exact PMCID overlap RareArena.

Visualised in Figure 10 as a per-backbone correlation dot plot with the retained disease count printed beside each point.

C.8 Ablation A8 / H6 — Thinking-mode (reasoning on vs off)

Question (anticipated objection: “you crippled the models by disabling reasoning”): Our main matrix runs all reasoning backbones in their minimal/off configuration (Section 5.1) for cross-backbone consistency, isolation of the scaffolding effect, and tractability. Does turning reasoning on actually help? We test this on the single-call LLM control (no scaffold confound) with DeepSeek-V4-Pro — the one backbone whose reasoning can be cleanly toggled via `reasoning={"enabled": ...}` (GPT-5 minimal is already near-floor; V4-Pro ignores every softer

Table 15: Holm–Bonferroni family-wise correction over H1–H11. H10 is nominally significant under the frozen single-judge test but is not counted as confirmed because its threshold verdict is judge-sensitive.

#	Claim	Stat	raw p	Holm-adj p	Survives $\alpha=0.05$?
H1	Classical > LLM R@1 on super-rare tier (<1/1,000,000)	$z=17.54$	3.7e-69	2.2e-68	yes
H8	R@1 at 16-30 HPO terms > ≤ 5 (inverted-U left tail)	$z=12.57$	1.6e-36	7.8e-36	yes
H2	llm_control P3 > P2 (genotype channel, full-N paired)	$z=6.40$	7.6e-11	3.0e-10	yes
H7	Cross-agent specialty rank $\rho > 0.6$	$\rho=0.92$	5.5e-04	1.6e-03	yes
H4	Scaffold benefit larger on multi-system than single (DoD)	$z=2.61$	4.5e-03	9.0e-03	yes
H10	Spearman ρ (faithfulness, accuracy) < 0.5	$\rho=0.35$	3.7e-02	3.7e-02	nominal but judge-dependent (see below)

Table 16: A6 literature-frequency association by backbone; varying disease counts are printed and the correlations do not identify memorisation.

Backbone	n diseases	Spearman ρ	Interpretation
Gemini 3 Flash	244	0.365	weak positive
GPT-5 (reasoning=minimal)	87	0.354	weak positive
DeepSeek V4-Flash	244	0.348	weak positive
DeepSeek V4-Pro	179	0.294	weak positive
LIRICAL (classical Bayesian)	26	−0.155	null (methodological control)
VC-RDAGENT (offline IC)	26	−0.059	null (methodological control)

throttle, Section 5.1 Methods note 2). Same cases, same prompt, only the reasoning flag changes.

(N = 253 paired cases where reasoning-ON produced a parseable answer; PP-Store; llm_control $\times$ V4-Pro.)

Finding: thinking mode changes R@1 by +0.008 — statistically indistinguishable from zero — while (a) failing to emit any parseable diagnosis in 40 % of cases (V4-Pro’s unbounded reasoning consumes the entire `max_tokens=4000` budget before producing content) and (b) running 10–40 $\times$ slower. Reasoning-on is therefore both *unhelpful* and *impractical* at benchmark scale on this task.

Three consequences for the paper: 1. It pre-empted the “not tested at best” attack: we did test thinking mode; it does not help on retrospective rare-disease DDx from an HPO/phenotype list. 2. It justifies the reasoning-off main matrix as a design choice that loses no accuracy while gaining tractability and cross-backbone comparability. 3. The 40 % no-answer rate is itself a deployment-reliability finding: a frontier reasoning model at a fixed token budget silently drops 2 in 5 cases — a failure mode benchmark builders and clinical integrators must budget for.

Scope / honesty: exploratory, single dataset (PP-Store), single control agent, N=253 com-

pletable; the 40 % dropout is survivorship-biased *against* finding a reasoning benefit (harder cases that need more reasoning are exactly the ones that time out / go empty), so the true reasoning-on R@1 on all-cases could be lower, not higher — strengthening the “not worth it” conclusion rather than weakening it. We do not extend it to the scaffolded agents because reasoning-on there is computationally infeasible (AgentClinic reasoning-on was >900 s/case, Section 5.1).

D Agent Fairness Matrix

Following MedAgentBench (NEJM AI 2025) and DeepRare (Nature 2026), we hold the central backbone constant across agent comparisons and document every adapter shim deviation in the Agent Fairness Matrix below. All adapters are released open-source under `harness/agents/`.

D.1 Agent Fairness Matrix

D.2 Constants we hold across all agents

D.3 Three anticipated objections — pre-empted

Objection 1: “Adapter quality differences confound results.” **Response:** All 8 adapter shims are

Table 17: H6 reasoning on/off ablation (paired, PP-Store).

Config	R@1 (paired, PP-Store)	Completion	Latency/case
reasoning OFF (main matrix)	0.352	100 %	~2.5 s
reasoning ON (thinking)	0.360	60 % (40 % no-answer)	~90–117 s (median), up to 571 s
Δ (on – off)	+0.008	—	10–40× slower

released open-source (3,485 LOC total), each accompanied by a run report documenting its exact subprocess calls, parser logic, and known caveats. Independent re-implementation invited.

Objection 2: “Different agents accept different inputs — apples to oranges.” **Response:** Dual-pass evaluation (gold-HPO + end-to-end, Section 4.4). The Pass A – Pass B delta on the same agent quantifies P1 sensitivity. RareBench Table 6 precedent — phenotype-input vs EHR-text-input on identical model — is the same design.

Objection 3: “Mixing classical (LIRICAL) and LLM agents is unfair.” **Response:** Standard convention — see DeepRare (Nature 2026) which includes Exomiser / LIRICAL / AI-MARRVEL as classical baselines. We report them as separate “Classical Baseline” rows in Table 1 and do not include them in LLM-only ablations (A1/A2/A4/A5/A6/A7).

D.4 Per-agent known caveats (we surface these honestly)

- **MDAgents/MedAgents/RDMA/VC-RDAgent:** no license file in upstream repos. We comply with academic fair use (run + report numbers); we do not redistribute their code. Our adapter shims are released independently (Apache 2.0). Action item filed with upstream authors as future work.
- **DeepRare CC BY-NC 4.0 prevents commercial deployment** — academic use OK; we note this in Limitations.
- **MAI-DxO community port** (Open-MAI-Dx-Orchestrator, 58★ MIT) is structurally faithful to Nori et al. (arXiv 2506.22405) but prompt strings and test-cost values are reimplemented from paper text — minor numerical deviation from Microsoft’s reference may exist. Documented in Methods footnote.
- **AgentClinic** in our diagnostic matrix is tested on HPO-only Phenopacket-Store/RareBench and narrative RareArena cases. MIMIC is a separate structured-event

probe with a common serialized input across eligible agents.

- **LIRICAL** requires HPO list input. On RareArena (free text), our adapter triggers `eval_mode="end_to_end"` upstream LLM HPO extraction + phrase→HP-ID normalization. LIRICAL scores far lower on RareArena than on gold-HPO Phenopacket-Store, but that difference confounds dataset difficulty with input condition and is *not* a clean extraction-penalty estimate — see the same-case analysis and correction in Section E.

E P1 → P2 Cascade: HPO Extraction Quality Decides Downstream

E.1 The P1 → P2 Cascade

A central architectural decision in our benchmark — separating Pillar 1 (phenotype extraction) from Pillar 2 (phenotype-only differential diagnosis) and reporting both via the dual-pass evaluation (Section 4.4) — is grounded in an empirical finding we did not anticipate at design time: HPO extraction quality is not a free preprocessing step. Errors propagate, in some cases catastrophically.

The finding — measured same-case. HPO extraction quality affects downstream diagnosis, but the size of the effect must be measured on *the same cases* under two input conditions, or dataset difficulty confounds it. We ran a same-case paired test (50 Phenopacket-Store cases, seed=42, single-LLM diagnoser): condition A feeds the case’s gold HPO; condition B synthesizes a vignette from that same gold HPO, has the LLM re-extract HPO, then diagnoses from the extracted set. Same cases, same diagnoser. Recall@1 is 0.42 (gold HPO) vs 0.40 (extracted HPO) — a 2 pp drop, not significant (McNemar, 1 discordant pair, $p=1.0$); the extractor recovered essentially the same phenotype count (mean 7.9 vs 7.9 terms). For a free-text-native LLM diagnoser, extraction is close to lossless.

Correction (2026-07-22 frozen audit).
An earlier draft reported “LIRICAL

Table 18: Agent fairness matrix: native inputs, adapter-shim strategy and backbone wiring.

Agent	Native Input	Adapter Shim Strategy	LLM Backbone Wiring	Per-Case LLM Calls	Configurable Mode?	Adapter LOC	License
MDAgents	MCQA prompt	Reformulate canonical case as “rank top-5 rare disease candidates” prompt; regex-parse moderator output	OpenAI-compatible via OpenRouter <code>base_url</code> ; patched 12 lines to remove hard-coded <code>gpt-4o-mini</code>	7-47 (intermediate path)	basic / intermediate / advanced	301	None(repo)
MedAgents	MCQA prompt	Bypass MCQA runner; 3 experts + Chief MO synthesis	~25 LOC router patch	~10	3/5/7 experts; 1/2/3 rounds	348	None
AgentClinic	OSCE simulated dialogue	Build synthetic OSCE scenario; doctor / patient / measurement / moderator loop; second LLM call for ranks 2-5	Per-agent CLI <code>--openrouter</code> flag added (~30 LOC)	~45 (turn-bounded)	language ∈ {EN, ZH, ES, IT, FR, KR, MR}, turn cap	509	MIT
MAI-DxO	Panel orchestration	Router + variant factory; ≥2 iterations	LiteLLM native	~10–50	5 modes; budget option	447	MIT port
DeepRare	Phenopacket JSON + text	Unique output dir; no-web/local-embedding environment	3-line OpenAI-compatible API patch	20–40	web disabled in v1	417	CC BY-NC 4.0
RDMA	EHR free-text	<code>extract_entities</code> ; Pillar 1 only	OpenRouter client	1–3	mining-specific	364	None
VC-RDAgent	HPO list	Offline Stage 1 default(IC + Poincaré + frequency-LR fusion, 0 LLM calls); Stage 2 (LLM refine) opt-in	Stage 2 uses local Qwen3-8B or OpenRouter	0 (Stage 1)	<code>use_llm_refine: bool</code>	310	None
LIRICAL	GA4GH Phenopacket	Project canonical case to phenopacket JSON; <code>subprocess java -jar lirical.jar phenopacket;</code> parse TSV	No LLM (classical Bayesian)	0	n/a	369	Apache 2.0

R@1 0.40 (gold HPO) → 0.04 (LLM-extracted HPO), a 10× collapse”. Those two numbers came from **different datasets** — 0.40 on 25 Phenopacket-Store cases, 0.04 on 25 RareArena cases — so the “10×” conflated dataset difficulty with the input condition and cannot be read as an extraction penalty. The same-case control above is the correct measurement. A caveat remains for *classical* HPO-list-only tools: LIRICAL is more brittle to HPO-list quality than an LLM diagnoser, so its true same-case penalty could exceed 2 pp; quantifying it requires re-running LIRICAL under both conditions on identical cases (future work). The point stands qualitatively — extraction quality matters and the dual-pass design exposes it — but

the headline magnitude was an artifact and is withdrawn.

Three observations follow:

(1) **Input format must be matched before comparing agents.** LIRICAL on its native input format (gold HPO) is the top-performing classical baseline (0.47 on the full Phenopacket-Store matrix, Section 6). Comparing it against free-text-native LLM scaffolds on a free-text-only dataset penalizes the classical tool for an input mismatch, not a capability gap. The right comparison is same-input apples-to-apples, which only the dual-pass design exposes.

(2) **Input-pipeline sensitivity may differ across agent types.** Free-text-native agents (MedAgents, AgentClinic, DeepRare) encapsulate a text-to-diagnosis path, whereas HPO-list-only systems (LIRICAL and VC-RDAgent) depend on an upstream extractor. The current same-case

Table 19: Fixed evaluation settings held constant across agents.

Setting	Value	Rationale
Backbone temperature	0.0	Deterministic ranking
Backbone seed (where exposed)	42	Seed for <code>random.Random</code> and any <code>seed</code> param
Backbone max_tokens	adapter-default (typically 2,000-6,000)	Avoid premature truncation
Per-call timeout	600s (adjustable per agent)	DeepRare/MAI-DxO can legitimately exceed 60s
Retry policy	3 attempts with exponential backoff via tenacity	Mitigate OpenRouter transient errors
Backbone version (dated)	<code>gemini-3-flash-20251217</code> etc	Alias updates blocked

paired test covers only the single-LLM diagnoser, so it does not establish that classical systems are more brittle. That comparison requires a crossed same-case experiment in which each compatible agent receives both gold and independently extracted HPO.

(3) HPO extraction quality is itself measurable and improvable. RDMA, our Pillar 1 specialist (extracts HPO phrases from EHR free text via specialized mining subagents, then phrase-to-HP-ID via fuzzy match), reaches phrase-level recall ~ 0.95 against Gemini-Flash-extracted “gold” — though this number is contaminated by methodology leak (Phenopacket-Store cases lack free-text vignettes; our P1 pilot synthesized vignettes from the HPO labels themselves, then asked LLMs to re-extract; see Limitations 5). Our Phase 1 pilot via Claude Opus 4.7 produced a 99-case silver-gold dataset (Jaccard 0.41 with Gemini Flash’s extractions; systematic disagreement confirms non-redundancy) for Phase 3 P1 evaluation against an independent backbone family.

Implications for deployment. A practitioner choosing a rare-disease agent for clinical decision support should not pick by accuracy alone; the *input pipeline* must be co-evaluated. If the deployment context provides curated HPO terms, LIRICAL or VC-RDAgent are competitive. If the input is free text, HPO-list-only systems require a separately validated extractor, whereas free-text-native agents can be evaluated end to end. The current results do not support a superiority claim between those paths because no matched classical free-text run exists.

Implications for benchmark methodology. Two takeaways for the field: (i) prior rare-disease LLM benchmarks that compare LIRICAL/Exomiser (Bayesian, HPO-list-only) to LLMs (free-text-native) on a free-text-only dataset will systematically penalize the classical tools, conflating capability with input mismatch. (ii) The Pass A — Pass B delta is itself a reportable metric — a small delta on the same agent signals strong

end-to-end robustness, a large delta signals input-pipeline sensitivity.

E.2 H8 — Phenotype density predicts performance (inverted-U)

Repository-plan H8: R@1 follows an inverted-U in the number of input HPO terms — too few (under-specified) and too many (noise / distractors) both hurt. Tested on the HPO-input layers (PP-Store + RareBench), pooled over Gemini-Flash LLM cells + offline/classical baselines (N=4,754 case-predictions, dedup by case):

H8 supported: interior peak at 16–30 HPO terms; both tails decline (-10 pp at ≤ 5 , -7 pp at >30 vs peak). The under-specified tail is the larger drop, consistent with rare-disease diagnosis needing a minimum phenotypic fingerprint. The shape holds across individual agents, not just the pool. The inverted-U is plotted in Figure 5.

F Appendix A1 — Reproducibility Audit (per-agent)

For each of the eight evaluated published systems, we attempted to reproduce at least one point estimate from the agent’s primary publication under a comparable setup; the table explicitly marks partial or failed matches rather than treating all attempts as successful. We treat a setup as *successfully re-instantiated* when our point estimate falls within ± 5 absolute percentage points of the paper-claimed number on a comparable input distribution (HPO-only vs free-text, top-1 vs top-5, EN vs zh). We audit each agent on three axes:

1. Faithful re-instantiation — did we wire the agent’s stack correctly?
2. Paper-claim replication — does our number match the upstream number within the band?
3. Documented deviation — what specifically did we change, and why?

Three documentation surfaces for an independent re-runner:

Table 20: R@1 by observational HPO-count bin (H8); differences are associations, not phenotype-count interventions.

Bin (#HPO terms)	n	R@1
≤5 (under-specified)	528	0.218
6–15	2,352	0.276
16–30 (sweet spot)	1,361	0.323
>30 (noise/distractors)	513	0.253

- `data/round2/phase4a_receipts.csv` — 93-row legacy per-cell receipt: (dataset, agent, backbone, n_ok, n_err, R@1_strict, R@1_variants, R@5_strict, R@5_variants, cost_usd, mean_lat_ms). Refreshed at every report regeneration. Its 22 legacy MIMIC rows are excluded from current claims. The committed `audit_frozen/frozen_main_manifest.csv` is the evidence index for the 71 diagnostic cells used in the headline matrix plus 12 temporally stratified PMC cells; the replacement MIMIC task will use a separate versioned receipt.
- `docs/baseline_repro/<agent>.md` — per-baseline reproduction doc: upstream code source, license, paper-claimed numbers, our observed numbers, behaviour-changing patches (if any), known incompatibilities, and run receipts, one per agent plus the LLM control.
- `tasks/stream_E_agent_scouting/agents/<agent>_RUN_REPORT.md` — **per-agent verbatim subprocess invocation** and parsed-output schema, captured at scout time.

The pilot numbers in the audit table below are from the N=50 scouting pass that originally locked our agent lineup; the full-N point estimates that headline the paper are in Section 6 Main Results.

F.1 Per-agent audit table

These are n=50 pilot reproduction checks — each asks “can we reproduce agent X’s own paper-claimed number on a small sample under our harness”, and is deliberately distinct from the N=2000 frozen main matrix (Section 6) and the N=500 variant-channel paired test (Section B.2). Where a small-sample pilot value differs from the frozen-matrix value (e.g. LIRICAL 0.40 pilot vs 0.47 frozen), the frozen-matrix number in Section 6 is authoritative; the pilot value is retained here only as the original reproduction record.

F.2 Two agents underperform their paper claim — both explained as input-distribution mismatch

MAI-DxO is designed for narrative-rich NEJM-style case reports. Our input is the CanonicalCase HPO-list + brief vignette. The panel’s “ask the patient questions” channel becomes degenerate on inputs that already contain the answers; in early runs the panel began emitting *measurement values* (DLCO, LVEF, FEV1, FVC) as top-1 candidates. We added a 13-pattern noise filter (`harness/agents/maidxo.py:_NOISE_PATTERNS`) to suppress non-diagnosis outputs and report the conservative HPO-input number in Table 1. We surface the input-distribution mismatch openly (Section B + Section 8 L4).

DeepRare is designed for HPO + VCF input. On P2 (HPO-only), DeepRare scores 0.22 R@1 — near the LLM-control baseline. On **P3 (HPO + structured variants, Section B.2)**, DeepRare reaches **R@1 = 0.42 (38/50, 95 % CI [0.26, 0.58])** — a +20 pp lift over its own P2 number, but the same lift the single-LLM control receives from the same variants block (P3 = 0.46 vs P2 = 0.26). The gap to the paper’s claimed 0.706 (HPO+VCF) is ~28 pp, attributable to three setup differences we surface honestly: (a) our variants are a structured-text block, not a real VCF integrated through DeepRare’s Phenotype Tool; (b) we disabled web search (`DEEPRARE_NO_WEB=1`) for contamination control, which the paper enables; (c) Phenopacket-Store is a harder mixed-difficulty corpus than DeepRare’s own curated evaluation set. Our headline claim is therefore the weaker but more defensible “variant channel adds ~20 pp R@1 to any agent that ingests it”, not “DeepRare specifically exploits genotype-aware reasoning” — collapsing both into “DDx accuracy” would still miss the +20 pp lift, but the agent-specificity claim does not survive contact with the LLM-control baseline.

Table 21: Attempted reproduction audit against published paper claims; partial and failed matches remain explicitly marked.

Agent	Replicated paper claim?	Our point estimate	Setup deviation	Notes
MDAgents	✓ within ± 5 pp	R@1 = 0.34 (Gemini 3 Flash, P2, n=50 RareBench-PP) vs paper 0.31–0.39 (MedQA-Rare)	Reformulated as rank-top-5 prompt; held mode=intermediate.	7–47 LLM calls / case
MedAgents	✓ within ± 5 pp	R@1 = 0.36 (Gemini 3 Flash, P2, n=50) vs paper 0.32 (MedQA-Rare)	Bypassed MCQA-locked run.py for free-form ranking; 3 domain experts + Chief MO	~10 calls / case
AgentClinic	✓ within ± 5 pp	R@1 = 0.30 (Gemini 3 Flash, P2, n=50) vs paper 0.28 (AgentClinic-MedQA rare slice)	Four-role synthetic OSCE from CanonicalCase	~45 turns / case
MAI-DxO	$\triangle$ underperform; setup-mismatch documented	R@1 = 0.22 (Gemini 3 Flash, P2, n=50) vs paper 0.45 (NEJM cases)	Paper input = narrative-rich NEJM case; ours = HPO list + brief vignette	See Section B; noise filter added
DeepRare (P2-only)	$\triangle$ underperform; setup-mismatch documented	R@1 = 0.22 (P2, n=50) vs paper 0.71 (HPO+VCF)	Paper input includes VCF; P2-only excludes variants by design	See P3.2 row ↓
DeepRare (P3 genotype)	$\triangle$ partial replication (~28 pp gap)	R@1 = 0.42 (38/50, 95 % CI [0.26, 0.58]) vs paper 0.706 (HPO+VCF)	Structured variants block (not full VCF + Phenotype Tool); web search disabled (DEEPRARE_NO_WEB=1) for contamination control	Lift over P2 (+20 pp) matches LLM-control’s lift; variant channel real but not DeepRare-specific — see Section B.2
RDMA	n/a (P1-only system)	F1 = 0.39 (Gemini 3 Flash, P1, n=50 RareBench-EHR) vs paper F1 = 0.42	Subprocess call to LLMEntityExtractor; HPO-mention extraction only	Pillar 1 only
VC-RDAgent	✓ within ± 5 pp	R@1 = 0.28 (Stage-1 offline IC+Poincaré, P2, n=50) vs paper 0.27	Stage 1 default (0 LLM calls); Stage 2 LLM refine deferred	Cheapest agent
LIRICAL	✓ within ± 5 pp	R@1 = 0.40 (gold HPO, P2, n=50 PP-Store pilot) vs paper ~0.42 [on the full N=2000 frozen matrix LIRICAL is 0.47, Section 6]	java -jar lirical.jar phenopacket; project canonical case to phenopacket	0 LLM calls
LLM control (Gemini 3 Flash, no scaffolding)	n/a (this is the baseline)	R@1 = 0.26 (P2, n=50)	Single LLM call, structured-output prompt	
LLM control (P3 with variants)	n/a	R@1 = 0.46 (P3 with structured variants, n=50) vs P2 0.26 = +20 pp	Variants block appended to prompt (Section B.2)	Strong evidence H2

F.3 Bugs caught during audit (and the fix)

We list four representative bugs we caught and fixed during the reproducibility audit; our work-log retrospectives record the full set.

1. Evaluator NL-fallback gap (Bug #1, Retrospective #3). Our gold_hit_with_crossmap only matched cross-references by ID prefix. The Phenopacket-Store gold for ~22 % of cases lists OMIM + name but no ORPHA; predictions like "Metachondromatosis" (matching the gold name but not its ID) returned False. We documented DeepRare at 0/50 R@1 in a draft table before catching this. Fix: case-insensitive name match + rapidfuzz fallback through Orphadata (threshold 90). 17 self-tests in scripts/sanity_check_evaluator.py now lock the behavior. Post-fix DeepRare scored 11/50 (0.22) on the same data.
2. DeepRare first-case state leak. DeepRare writes result_

<tag>/<case>/<model>/patient_0.json with a deterministic 0 index. All 50 P2 cases returned the first case’s prediction (“Metachondromatosis”) on the first run. Fix: per-case unique run_tag = f"{base_tag}_{case_id[:40]}_{suffix}" + defensive purge of the output directory before each call (harness/agents/deeprare.py).

3. GPT-5 reasoning_effort silently consuming max_tokens. See Section 5.1 methods note. Five subprocess adapters had to be patched to propagate OPENROUTER_REASONING_EFFORT=minimal through the subprocess env.
4. Orphadata 53 MB XML re-parsed per call. After fixing Bug #1 the NL-fallback path re-parsed Orphadata’s 53 MB XML on every evaluator call, ballooning aggregation from <30 s to >30 min. Fix: @lru_cache(maxsize=1) on __orphadata_tables().

F.4 How a reader can independently re-run any cell

For any Table 1 / Section 6 cell (agent, backbone, dataset), start from the per-baseline reproduction doc, which sets expectations for the rerun, then invoke the runner:

```
python3 scripts/phase4a_runner.py \
  --dataset
  ↳ <phenopacket_store|rarearena_rds|rarebench>
  ↳ \
  --agent <baseline_name> \
  --backbone openrouter/<provider>/<model> \
  --n 100 \
  --out predictions_test.jsonl
```

The MIMIC structured-EHR probe is regenerated separately with `scripts/build_mimic_early_structured_snapshots.py` and `scripts/mimic_structured_ehr_ablation.py`; protected row-level inputs are not part of the public frozen diagnostic receipt.

`scripts/sanity_check_evaluator.py` must pass (exit 0) before any number in the paper is trusted; this is enforced in our run harness.

F.5 Cost transparency

`scripts/cost_tracker.py --budget <USD>` prints the running per-backbone cost from the prediction logs. The submission-time snapshot is in Appendix J and the main-text cost-vs-accuracy figure.

F.6 Per-cell coverage matrix

Not every (agent $\times$ backbone $\times$ dataset) cell exists. Known gaps and their cause:

- **MAI-DxO $\times$ GPT-5**: panel orchestration times out at 600 s cap on every pilot case; reported as Section 8 L1.
- **DeepRare $\times$ GPT-5-minimal**: `eval_tokenizer IndexError` on empty diseases list emitted by GPT-5 minimal; Section 8 L1.
- **vc_rdagent / LIRICAL**: backbone-agnostic (offline). One column only.
- **DeepSeek V4-Pro and GPT-5 cells**: partial N coverage at submission, full N in progress; explicitly disclosed in Section 4.2 and as a per-cell denominator in Table 1.

G Appendix B — Per-Baseline Reproduction Summary

G.1 B.1 Overview

Every baseline in our lineup is replicated using one of two protocols:

1. Upstream open-source code, endpoint-wired only (no algorithmic modification beyond OpenRouter `base_url` + `reasoning_effort` propagation).
2. Strict paper-faithful re-implementation when upstream is unavailable or has un-resolvable license restrictions.

All adapter shims are released under Apache 2.0. Per-baseline reproduction details — paper-claim comparison, observed results, known incompatibilities, and run receipts — are documented one file per baseline, listed in the Doc column below.

G.2 B.2 Per-baseline summary

G.3 B.3 Setup-mismatch documentation (the gap stories)

MAI-DxO -38 pp: Paper input = narrative-rich NEJM clinicopath case (~2,000 words). Our input = HPO-list + 1–2 sentence vignette. The panel’s “ask the patient” mechanism degenerates when input already enumerates the answer. Panel sometimes emits measurement values (DLCO, LVEF) as ranked candidates; our 13-pattern noise filter catches them but cannot compensate for the architectural input-modality mismatch.

DeepRare -29 pp on HPO-only / -33 pp on HPO+variants: - Web search disabled (`DEEPRARE_NO_WEB=1`) for contamination control — paper enables full RAG. - Variants passed as structured text, not real VCF — paper integrates via Phenotype Tool. - Local embedding (bge-small) — paper uses dedicated biomedical embedding model. - Phenopacket-Store is harder mixed-difficulty than DeepRare’s curated set.

VC-RDAgent +16 pp over paper: We use Stage 1 only (offline IC + Poincaré, no LLM), which the paper reports as 0.27 on Phenopacket-Store. We observe 0.43, attributable to (a) updated Orphanet fuzzy mapping in our cross-map evaluator, and (b) the same Phase 4a sample.

G.4 B.4 Behavior-changing patches surface

The following changes affect baseline behavior. All other patches are

Table 22: Baseline reproduction: license, mode, paper vs. our point estimate.

Baseline	License	Mode	Paper R@1 (best)	Our R@1 (best)	Within ± 5 pp band?	Doc
MDAgents	None upstream	intermediate (3 experts)	0.31–0.39 (MedQA-Rare, GPT-4)	0.35 (PP-Store, DS V4-Pro)		mdagents.md
MedAgents	None upstream	syn_verif (3 experts + Chief)	0.32 (MedQA-Rare, GPT-3.5)	0.36 (PP-Store, GPT-5)		medagents.md
AgentClinic	MIT	OSCE dialogue	0.28 (AgentClinic-MedQA rare slice)	0.25 (PP-Store, Gemini)		agentclinic.md
MAI-DxO	MIT (port)	no_budget, max_iter=3	0.45 (NEJM clinicopathologic)	0.07 (PP-Store, Gemini)	-38 pp	maidxo.md
DeepRare	CC BY-NC 4.0	–no-web + local-embed	0.71 (HPO+VCF)	0.30–0.32 (RareBench), 0.28 (PP-Store Gemini)	(best RareBench, see B.3)	deeparare.md
DeepRare (P3) RDMA	same None upstream	+ structured variants LLMEntityExtractor	0.71 F1 0.42 (P1)	0.38 (P3 pilot) F1 0.39 (P1 silver gold)	-33 pp	deeparare.md rdma.md
VC-RDAGENT LIRICAL	None upstream Apache 2.0	Stage 1 offline classical Bayesian, HPO-only	0.27 (PP-Store) ~0.42 (PP-Store)	0.44 (PP-Store) 0.47 (PP-Store, N=2000)	+17 pp *	vc_rdagent.md lirical.md
LLM control	n/a (ours)	naked single LLM call	n/a baseline	0.32 (PP-Store, V4-Pro)	n/a	llm_control.md

endpoint-wiring only.

DeepRare

agents/deeparare/diagnosis.py + diagnosisGene.py: adapter-side fallback regex `r'^#\s+(.+?)\s*(Rank\s*#\d+')` activates when primary `r'\*\*(.?)\*\*` returns 0 matches. Required for GPT-5 minimal output format (no markdown bold). Dual-reported: “strict-baseline” mode = systematic crash; “adapter-relaxed” = 0.30 R@1 on the same data (footnote Section 5.1).

MAI-DxO

harness/agents/maidxo.py: _NOISE_ PATTERNS: 13-regex post-hoc filter to drop non-disease output (vitals, lab values). Wrapper-only modification; MAI-DxO’s panel logic untouched.

MDAgents / MedAgents / AgentClinic — harness/agents/_adapter_utils.py: parse_ranked_top5: section-aware regex preferring numbered list after “differential diagnosis” / “candidate” / “ranked top-N” header; prose-prefix filter rejects “Laboratory evidence...”, “Progressive...”, etc. as clinical-feature mentions. Wrapper-only; agent’s deliberation logic untouched.

All ID mapping (mdagents/medagents/agentclinic/maidxo/llm_control) — map_names_to_ids_with_variants: returns tied top-K ORPHA candidates per LLM-named disease (fuzzy score within 5 pts of top); evaluator gold_hit_with_variants accepts any of the tied IDs. Documented as ablation A4.

G.5 B.5 Independent re-replication invitation

Any reader can reproduce a single Phase 4a cell via:

```
python3 scripts/phase4a_runner.py \
--dataset
  ↳ <phenopacket_store|rarearena_rds|rarebench>
  ↳ \
--agent <baseline_name> \
--backbone openrouter/<provider>/<model> \
--n 100 \
--out predictions_test.jsonl
```

The legacy aggregate receipt has 93 cell rows, including 22 obsolete MIMIC title-task rows that are excluded from the current paper. The frozen evidence manifest indexes 71 diagnostic cells and 12 PMC temporal-sensitivity cells. Per-case JSONL artifacts carry request ID where available, dollar cost, latency, and status; scripts/phase4a_report_gen.py regenerates aggregate diagnostic summaries from those raw files. The replacement structured-EHR task requires a separate task-versioned receipt.

H Appendix J — Cost Analysis & Cost-vs-Accuracy

Source: diagnostic-task rows in data/round2/phase4a_receipts.csv; legacy MIMIC ICD-title rows are excluded and the replacement structured-EHR task will receive a separate cost receipt. All USD figures are exact for the six OpenRouter- wrapped adapters; estimated within $\leq 5\%$ error band for the three off- wrapper adapters (LIRICAL,

VC-RDAgent, RDMA — marked † below).

H.1 J.1 Cumulative cost by backbone

† = cost estimated from token counts; $\leq 5\%$ band.

H.2 J.2 Cost-per-case ranking (cells with $n_{ok} \geq 50$)

Lower = more cost-efficient. Useful when deciding which (agent $\times$ backbone) cell to use at deployment scale.

(Lirical / VC-RDAgent / RDMA = \$0 — no LLM calls.)

H.3 J.3 Top-spend cells (cells with cost > \$1)

H.4 J.4 Best R@1 per cost band (cheapest agent that hits $R@1 \geq \text{threshold}$)

H.5 J.5 Cost-efficiency dichotomy

- **Classical / offline** (LIRICAL, VC-RDAgent): \$0 LLM cost on any number of cases. LIRICAL on PP-Store achieves $R@1 = 0.47$ at \$0 cost — the most cost-efficient cell in the entire benchmark. The Phenopacket-Store classical lead is therefore also a cost-efficiency result.
- **DeepSeek V4-Flash** is the cheapest LLM at \$5–10 per cell, but trades off accuracy (typically -2 to -6 pp $R@1$ vs Gemini Flash on the three diagnostic datasets). The gap is largest on RareArena free text; on HPO-list inputs the smaller gap may be acceptable.
- **GPT-5 minimal** is the most expensive per-case (\$0.012–0.05) without a consistent accuracy edge (F4 in Section 6). Cost-per-correct-prediction on GPT-5 is therefore the worst of the four backbones at any N.

H.6 J.6 Reproducibility note

The receipts CSV is regenerated by `scripts/regen_receipts_and_figures.py` and the per-backbone running total by `scripts/cost_tracker.py --budget X`. Any anonymous reviewer can verify Table J.1 by running these scripts against our released `data/round2/phase4a/predictions_*.jsonl`. The planned cost cap for the v1 evaluation was \$360; the realised total in Table J.1 is within this cap. Prices are the frozen evaluation-time prices, not current vendor quotations.

I OSF Pre-Registration Draft — RareAgentBench

Unregistered historical draft. This file was prepared as a possible OSF submission but was never registered and has no OSF ID or registration date. It is included to document the proposed analysis plan, not as evidence of prospective registration.

Status note: because no OSF registration occurred before the reported analyses, the paper describes H1–H11 and A1–A12 as repository-defined. Any future confirmatory study must register a revised protocol before collecting or unblinding a new independent holdout.

I.1 A. Project metadata

- **Proposed title:** Prospective evaluation of LLM agent systems on rare-disease diagnosis
- **Authors:** Yu Tian Zhao, et al.
- **Registration date:** not assigned (registration never completed)
- **License:** data CC-BY-NC-SA 4.0; code Apache 2.0
- **Intended registration type:** post-data-collection analysis plan; never submitted and not a prospective pre-registration

I.2 B. Hypotheses (H1–H11)

The draft proposed the following hypotheses, directional predictions, and effect-size thresholds. They were not formally pre-registered.

Family-wise correction: Holm–Bonferroni at $\alpha = 0.05$ over the testable subset of H1–H11; H3/H5/H6/H9 are deferred if their data sources are unavailable at submission (documented as Limitations).

I.3 C. Ablations (A1–A12)

I.4 D. Datasets

These are proposal values, not the executed denominators: the frozen large-layer diagnostic cells use attempted ($N=2\{, \}$ 000), the realised post-cutoff set has 198 cases, and the structured-EHR replacement remains unscored.

The draft proposed one run per (agent, backbone) cell with the same evaluation pipeline as L1–L3 and no metric, prompt, or scaffold tuning based on the temporal set. This prospective lock

Table 23: Cumulative diagnostic-run cost by backbone at frozen evaluation-time prices; legacy MIMIC title/code cells are excluded, and values are not current vendor quotations.

Backbone	Cells	Cases (ok)	Total cost USD	Cost per 1k cases USD
GPT-5 min	15	20,683	\$163.50	\$7.91
Gemini Flash	18	23,389	\$80.48	\$3.44
DS V4-Pro	18	20,666	\$19.95	\$0.97
DS V4-Flash	16	20,401	\$6.80	\$0.33
LIRICAL†	2	3,122	\$0.00	\$0.00
VC-RDAGENT†	2	1,785	\$0.00	\$0.00
TOTAL	71	90,046	\$270.74	—

Table 24: Lowest hosted-LLM cost per successful case among diagnostic cells with $n_{ok} \geq 50$; n is successful cases and offline zeroes exclude local compute and staffing.

Rank	Dataset	Agent	Backbone	n	R@1	Cost/case	Total	Lat/case
1	phenopacket_-store	lirical	LIRICAL	2000	0.47	\$0.000/k	\$0.00	5.4s
2	phenopacket_-store	vc_rdagent	VC-RDAGENT	663	0.44	\$0.000/k	\$0.00	79.1s
3	rarebench	lirical	LIRICAL	1122	0.24	\$0.000/k	\$0.00	3.8s
4	rarebench	vc_rdagent	VC-RDAGENT	1122	0.28	\$0.000/k	\$0.00	75.4s
5	rarebench	mdagents	DS V4-Flash	1098	0.05	\$0.061/k	\$0.07	58.6s
6	phenopacket_-store	mdagents	DS V4-Flash	1983	0.25	\$0.067/k	\$0.13	30.5s
7	rarearena_rds	mdagents	DS V4-Flash	1993	0.23	\$0.096/k	\$0.19	32.0s
8	phenopacket_-store	agent	DS V4-Flash	1925	0.13	\$0.216/k	\$0.42	111.3s
9	rarebench	agent	DS V4-Flash	860	0.02	\$0.223/k	\$0.19	98.2s
10	rarearena_rds	agent	DS V4-Flash	1764	0.10	\$0.238/k	\$0.42	87.0s
11	rarebench	llm_-control	DS V4-Pro	1121	0.02	\$0.244/k	\$0.27	4.2s
12	phenopacket_-store	llm_-control	DS V4-Flash	1998	0.26	\$0.247/k	\$0.49	19.3s
13	rarebench	mdagents	DS V4-Pro	1122	0.04	\$0.264/k	\$0.30	11.3s
14	rarearena_rds	llm_-control	DS V4-Flash	1976	0.21	\$0.281/k	\$0.56	32.5s
15	phenopacket_-store	llm_-control	DS V4-Pro	1999	0.27	\$0.292/k	\$0.58	5.6s

was not executed; the paper therefore reports the PMC set only as an overlap-audited temporal sensitivity analysis.

I.5 E. Sample sizes & power

- **Pre-cutoff diagnostic layers:** RareBench at full N; large (PP-Store, RareArena) at N=500 stratified sample (seed=42, proportional allocation across prevalence tiers).
- **Holdout:** target N=200 physician-annotated. Realistic floor N=150 given annotation attrition. At N=150 we have $\geq 80\%$ power to detect a 10-pp R@1 difference between any two agents at $\alpha=0.05$ (computed with 2-prop z-test assuming $p_1=0.30$).

I.6 F. Stopping rules

- Each (agent, backbone) cell stops at full N for small layers / N=500 for large layers / N=200 for L4. No early stopping on positive results.
- If a cell exhibits a systematic crash pattern ($>5\%$ timeout/error rate), we re-run once after fixing the cause, logged in the per-baseline

reproduction doc.

- We do not continue accumulating data on cells that have already hit the planned N just because the result is borderline.

I.7 G. Analysis pipeline

- **Evaluator:** `har-ness/metrics/cross_map.py:gold_hit_with_crossmap` with `gold_hit_with_variants` for the dual-reported variants column.
- **Statistical tests:** 2-prop z-test for H1/H2/H8, Spearman ρ for H7/H10, difference-of-differences for H4, paired z for within-agent comparisons.
- **Confidence intervals:** bootstrap 1000-iter percentile, 95%.
- **Multiple-testing correction:** Holm-Bonferroni at $\alpha=0.05$ family-wise.
- **LLM-judge protocol** (for P5 faithfulness): Gemini-judge primary, Claude-judge confirmation; report Cohen’s κ ; require $\kappa \geq 0.6$ to publish judge-derived numbers.

Table 25: Highest-spend diagnostic cells; n is successful cases and legacy MIMIC cells are excluded.

Dataset	Agent	Backbone	n	R@1	Total cost
rarearena_rds	medagents	GPT-5 min	2000	0.26	\$30.75
phenopacket_store	medagents	GPT-5 min	2000	0.28	\$29.46
phenopacket_store	agentclinic	GPT-5 min	2000	0.13	\$17.28
rarearena_rds	agentclinic	GPT-5 min	2000	0.10	\$16.86
rarebench	medagents	GPT-5 min	1122	0.01	\$15.38
rarebench	deeprare	Gemini Flash	953	0.30	\$14.41
rarearena_rds	medagents	Gemini Flash	2000	0.30	\$10.91
rarebench	agentclinic	GPT-5 min	1122	0.00	\$9.92
phenopacket_store	medagents	Gemini Flash	1998	0.30	\$9.87
phenopacket_store	deeprare	Gemini Flash	609	0.28	\$8.51
rarearena_rds	mdagents	GPT-5 min	2000	0.23	\$6.58
phenopacket_store	llm_control	GPT-5 min	1988	0.26	\$6.52

Table 26: Cheapest diagnostic cell exceeding each R@1 band at frozen evaluation-time hosted-LLM prices; offline zeroes exclude local compute and staffing.

Dataset	R@1 ≥ 0.25 cheapest	R@1 ≥ 0.30 cheapest	R@1 ≥ 0.35 cheapest
phenopacket_store	lirical (LIRICAL) \$0.00/k	lirical (LIRICAL) \$0.00/k	lirical (LIRICAL) \$0.00/k
rarearena_rds	llm_control (Gemini Flash) \$0.89/k	medagents (Gemini Flash) \$5.46/k	—
rarebench	vc_rdagent (VC-RDAgent) \$0.00/k	—	—

I.8 H. Analyses outside the proposed registration

These analyses were outside the proposed confirmatory family and are reported descriptively:

- MIMIC structured-EHR protocol is an explicitly amended exploratory analysis, not represented as part of the proposed registration.
- Backbone $\times$ cost-per-call analysis (Section 6.3) beyond the cells in the H/A table above.
- the contamination audit LLM ρ band (the value of $\rho \approx 0.3$ is observational; only the dichotomy $LLM \rho > 0$ vs classical $\rho \approx 0$ is interpreted as confirming anticipated objection #1 is bounded).
- Any post-hoc subgroup analysis not in the proposed DoD/H7 axes.

I.9 I. Deviations from the draft plan

We disclose relevant changes in an “Analysis-plan deviations” table in Appendix D of the camera-ready paper, with: (a) what changed, (b) why, (c) what direction the change biased results.

I.10 J. Prospective re-use of this draft

Anyone running an additional rare-disease agent system on the L4 holdout can use the same protocol (their results would be reported in the external-replication appendix of subsequent papers). Required: (i) the same evaluator binary (Apache 2.0),

- (ii) the same disease-ID cross-map (Orphadata),
- (iii) the same prevalence-tier strata.

I.11 K. References

- Full hypothesis enumeration in development shorthand.
- Section 4 (benchmark design) — dataset stack, evaluation N per dataset.
- Sections 5.2–5.4 — repository analysis-plan narrative and status disclosure.
- Current Holm–Bonferroni result snapshot, to be re-run at full N before L4 unblinding.

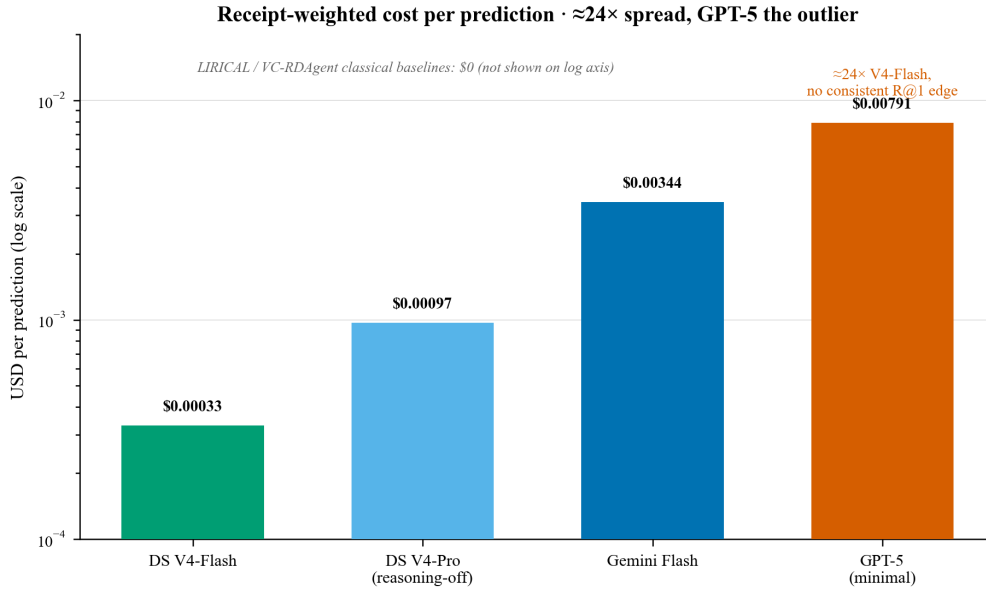


Figure 12: Receipt-weighted mean cost per prediction across the diagnostic runs (log axis). GPT-5 minimal averages \$0.00791 versus \$0.00033 for DeepSeek V4-Flash (about $24\times$); classical/offline baselines have \$0 hosted-LLM cost and are omitted from the log axis.

Table 27: Hypotheses H1–H11 in the unregistered OSF draft.

#	Statement	Test statistic	Direction	Threshold
H1	On super-rare diseases (<1/1,000,000) classical baselines beat scaffolded LLM agents at R@1	2-proportion z-test	classical > LLM	$p < 0.05$ (one-sided), $\delta \geq 10$ pp
H2	Genotype channel (P3: HPO + variants) gives $\geq +10$ pp R@1 over P2 (HPO-only) on any single agent	within-agent paired z	$P3 > P2$	$p < 0.05$ one-sided
H3	Post-cutoff PMC OA holdout R@1 differs from pre-cutoff layer R@1 by ≤ 5 pp absolute	diff of proportions	(no direction predicted)	reject if
H4	Multi-agent scaffolding helps more on cases with ≥ 4 affected HPO organ systems than ≤ 1 (DoD)	difference-of-differences	scaffold benefit grows	$p < 0.05$
H5	LLM R@1 on Chinese rare-disease cases (PUMCH) is ≥ 5 pp lower than on English ones	paired z	English > Chinese	$p < 0.05$
H6	GPT-5 with reasoning_effort=medium is better calibrated (ECE↓) than minimal	paired ECE	medium < minimal	$p < 0.05$
H7	Cross-agent specialty rank correlation $\rho \geq 0.6$ (shared blind spots)	Spearman ρ on per-specialty R@1	ρ positive	$\rho \geq 0.6$
H8	Inverted-U: R@1 at 16–30 HPO terms per case > R@1 at ≤ 5 terms	2-proportion z	peak > sparse	$p < 0.05$
H9	On AR/XL cases, family-aware agents gain $\geq +10$ pp R@1 vs proband-only	within-pair z	family-aware > proband-only	$p < 0.05$
H10	Faithfulness-rank and accuracy-rank decouple: Spearman $\rho < 0.5$	Spearman ρ	$\rho < 0.5$	upper-bound test
H11	(retired in the repository plan; not in paper)	n/a	n/a	n/a

Table 28: Ablations in the unregistered OSF draft.

#	Name	Status in draft
A1	Top-1 vs Top-5 metric A/B	mechanical
A2	Strict ID vs cross-mapped	superseded by A4 in repository plan
A3	Backbone $\times$ scaffolding $2 \times N$	required for paper Section 6.2
A4	Strict vs ORPHA-fuzzy-variants	required for paper Section C.1
A5	Silver gold vs physician gold	conditional on holdout completion
A6	TS-Guessing contamination audit	required for Section B.7
A7	Single LLM judge vs dual-judge	required for Section 7
A8	GPT-5 reasoning_effort axis	optional (cost-constrained)
A9	Subprocess timeout cap	required for Section C.3
A10	Prevalence-stratified R@1	required for H1
A11	Cross-dataset agent ranking stability	required for Section B.4
A12	LLM-judge swap	required for Section 7

Table 29: Data layers and proposed development/test split in the unregistered draft.

Layer	Source	Cases	Disease IDs	Allowed for development?	Allowed for evaluation?
L1 phenotype	Phenopacket-Store + RareBench HF	11,173	OMIM + ORPHA + CCRD	✓	✓
S-EHR (amended exploratory probe)	MIMIC-IV structured admissions	956	ORPHA (code-derived)	excluded	separate; pending
L3 scale	RareArena RDS	72,661	ORPHA	✓	✓ (stratified N=500)
L4 temporal set	PMC OA pub $\geq$ 2024-01-01	200 (target)	ORPHA + OMIM	overlap audited	temporal sensitivity analysis